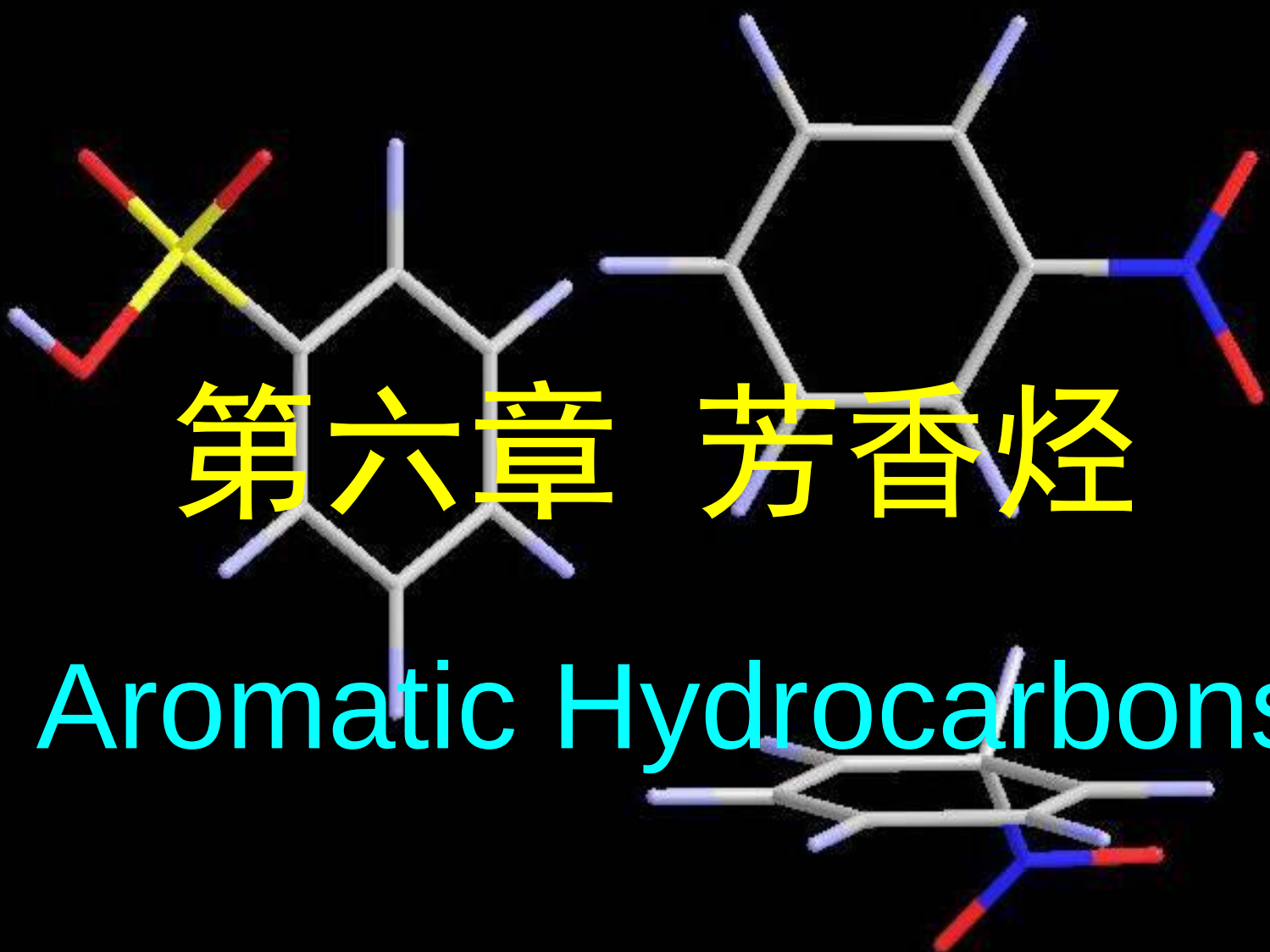


The background features several chemical structures. On the left, a benzene ring is substituted with a sulfonamide group (-SO<sub>2</sub>NH<sub>2</sub>) and a methyl group. On the right, a benzene ring is substituted with a methyl group and a dimethylamino group (-N(CH<sub>3</sub>)<sub>2</sub>).

# 第六章 芳香烃

Aromatic Hydrocarbons

# CONTENT

- 1 芳香烃的分类、结构及命名
- 2 芳香烃的物理性质(自学)
- 3 芳香烃的化学性质
- 4 苯环上亲电取代的定位规律
- 5 稠环芳烃

# 何为芳香烃？

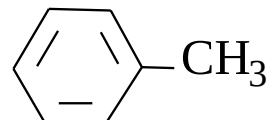
- 含有苯环结构的烃称为芳香烃, 简称芳烃 (aromatic hydrocarbon)。芳香烃的名称早期发现来源于香树脂、香精油等具有芳香气味的物质。
- 具有芳香族化合物性质, 但没有苯环结构的化合物称为非苯芳烃 (nonbenzenoid aromatic hydrocarbon)。

# 芳香烃的分类、结构及命名

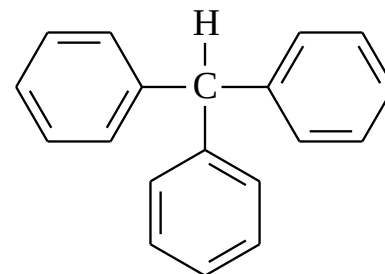
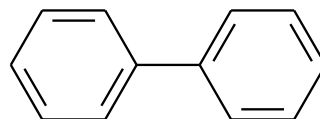
## 芳香烃的分类

### 芳香烃

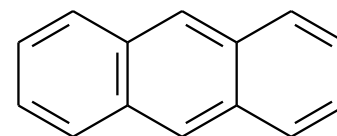
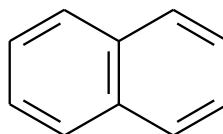
单环芳烃



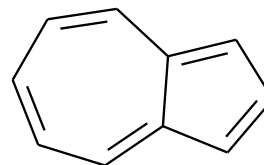
多环芳烃



稠环芳烃



非苯芳烃



不含苯环，但具有苯的特性“芳香性”的碳环化合物

# 苯的结构

## 苯的特征:

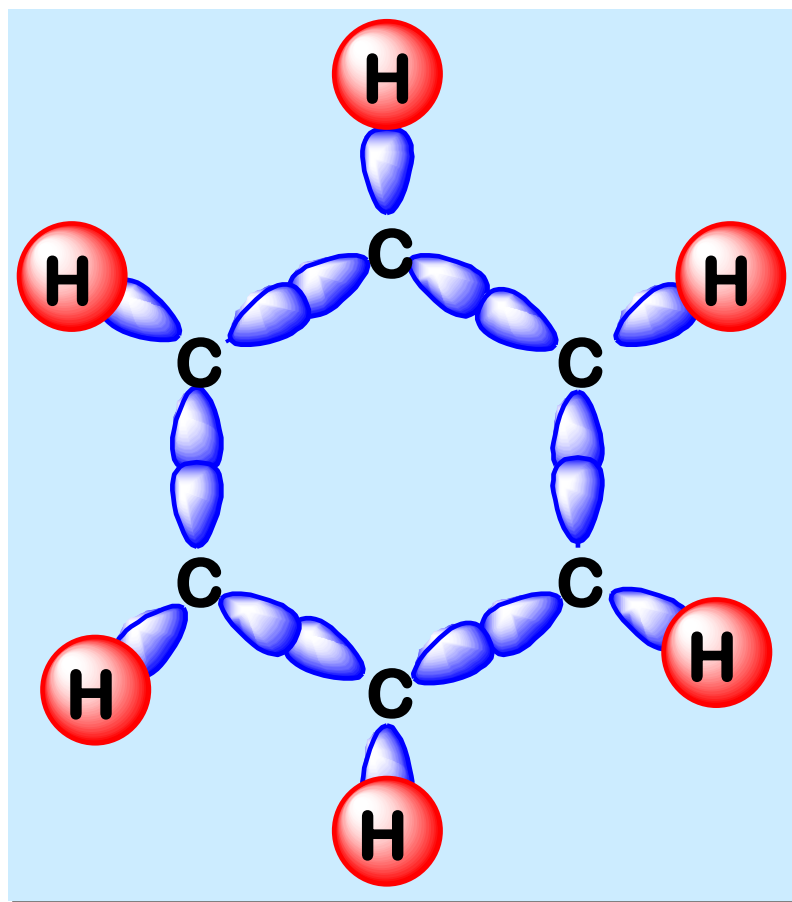
- 分子式 $C_6H_6$ ，C:H=1:1，不饱和度为4
- 具有特殊稳定性和对化学反应的惰性

## 芳香族化合物的特性 —— 芳香性

- 苯环易发生取代，难发生加成反应
- 不易氧化
- 苯环具有特殊稳定性

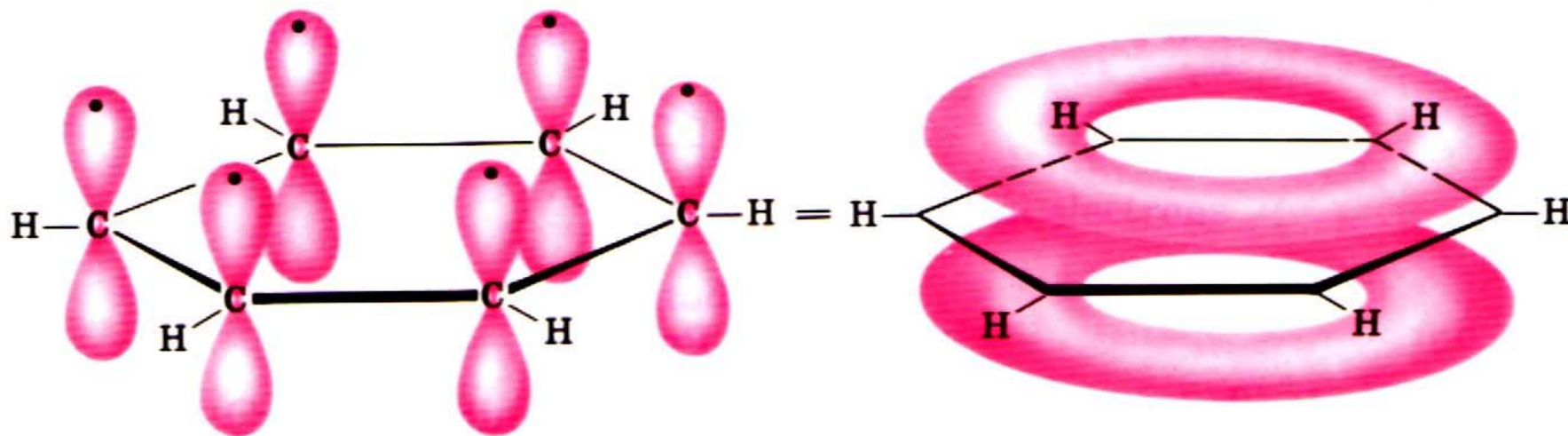
# 苯的分子轨道描述

- 6个碳原子都是 $sp^2$ 杂化



# 苯的分子轨道描述

- 未参与杂化的 $p$ 轨道带着一个电子垂直于环平面，形成一个闭合环状的大 $\pi$ 键

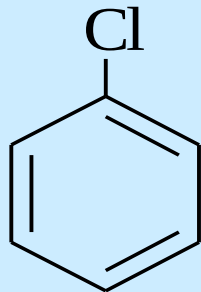


- 苯的6个碳原子形成平面正六边形
- 碳-碳键长均等，为  $1.40 \text{ \AA}$
- 所有键角均为  $120^\circ$

# 苯及其衍生物的命名和异构

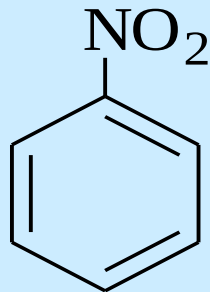
## 1. 单取代苯的命名

苯环作为母体： - X, - NO<sub>2</sub>, - NO, 简单烷基



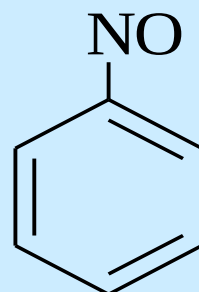
氯苯

Chlorobenzene



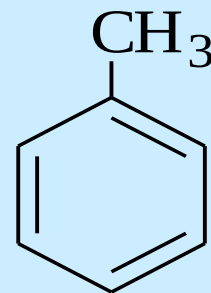
硝基苯

nitrobenzene



亚硝基苯

nitrosobenzene

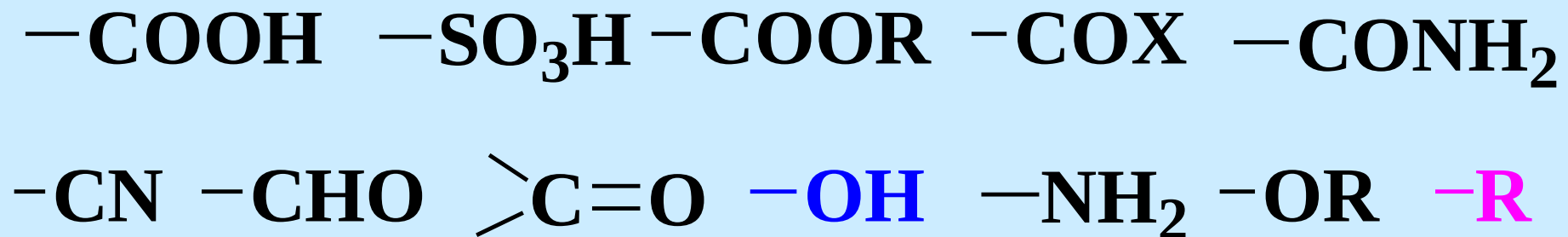


甲苯

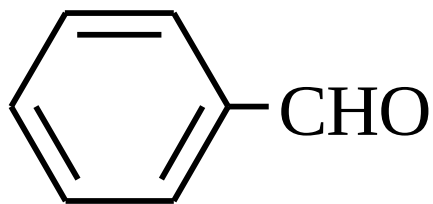
methylbenzene  
(Toluene)



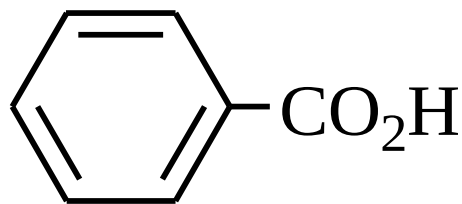
# 苯环作为取代基： 1) **R**之前基团与苯环相连时



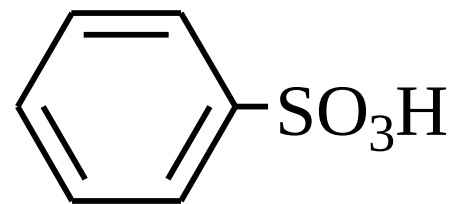
- OH, - NH<sub>2</sub>与苯环一起构成母体 “苯酚”、  
“苯胺”



苯甲醛



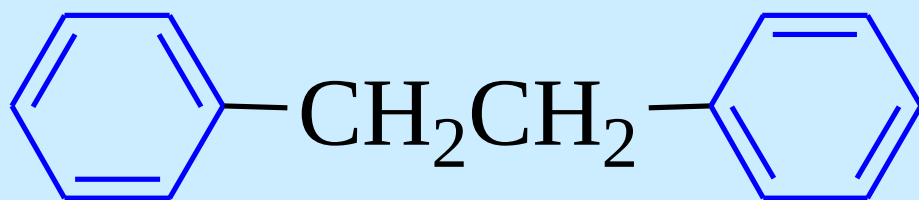
苯甲酸



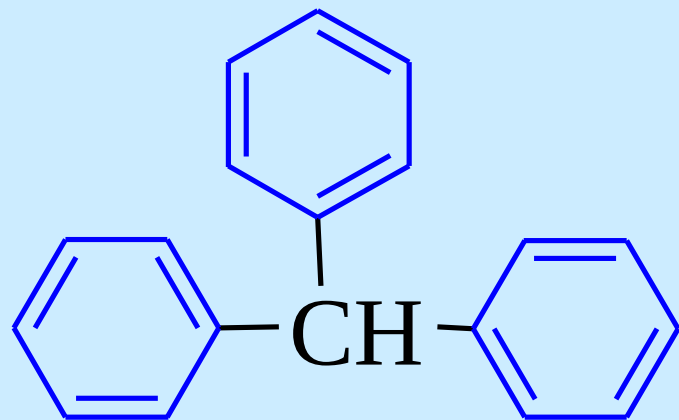
苯磺酸

# 苯环作为取代基

2)分子中有多个苯环，  
以任一苯环为母体命名都有困难时



1,2-二苯基乙烷

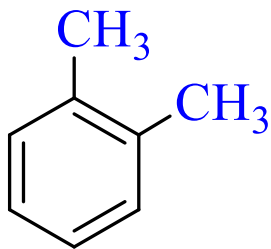


三苯基甲烷

# 二取代苯的命名

- 2个取代基相同

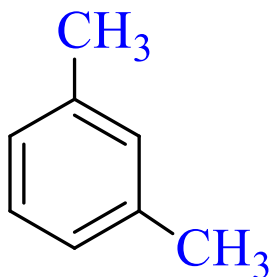
(1,2-*o*-邻; 1,3-*m*-间; 1,4-*p*-对)



1,2-Dimethylbenzene  
(*o*-Xylene)

*ortho*

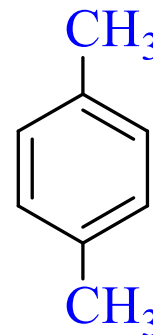
邻-二甲苯



1,3-Dimethylbenzene  
(*meta*-Xylene)

*meta*

间-二甲苯



1,4-Dimethylbenzene  
(*para*-Xylene)

*para*

对-二甲苯

# 芳香烃的化学性质

芳香亲电取代反应

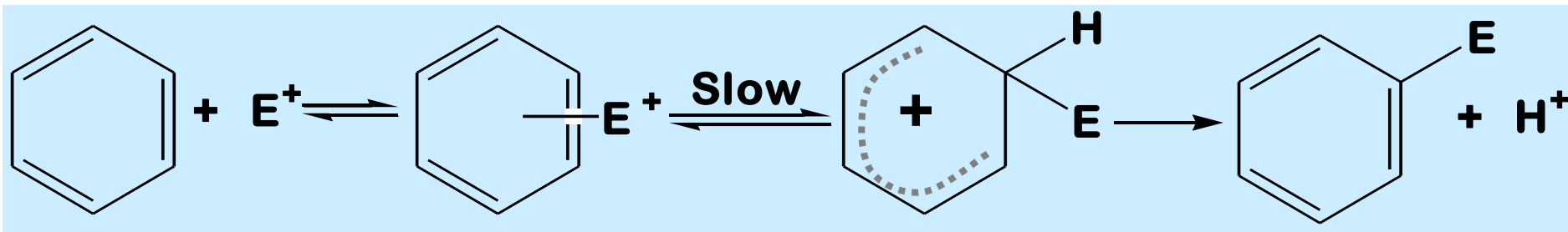
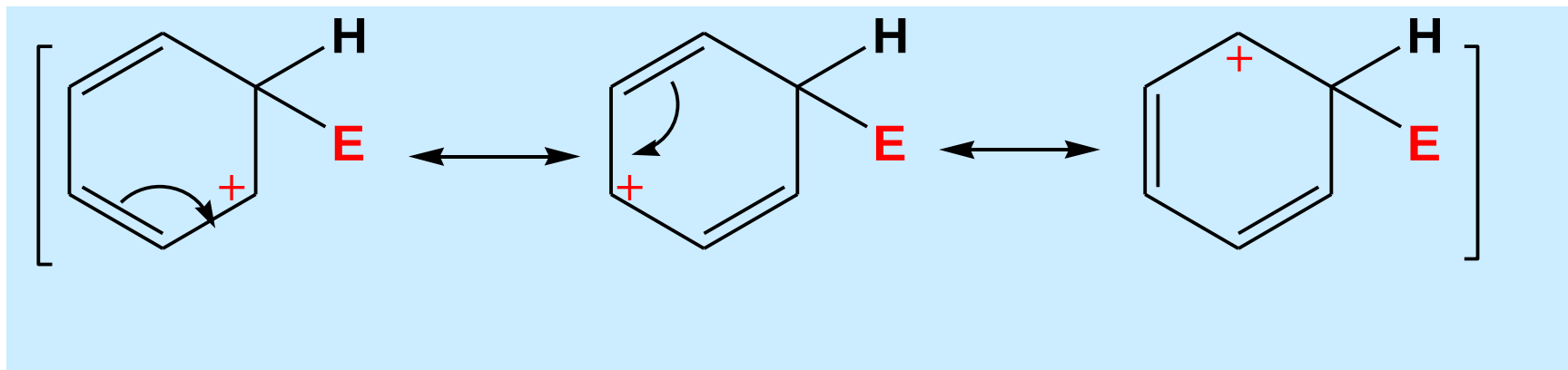
加成反应

氧化反应

$\alpha$ -卤代反应

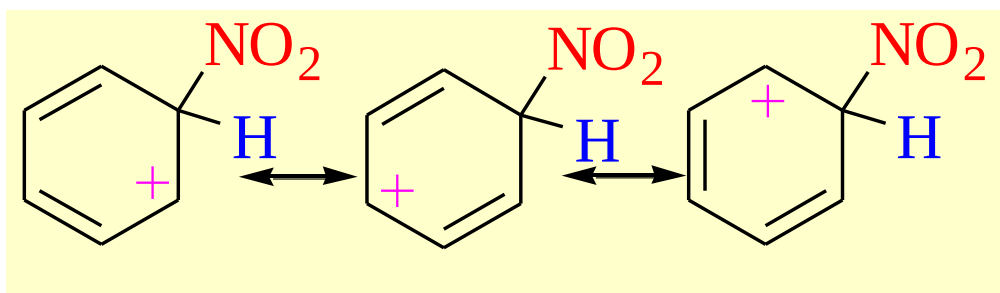
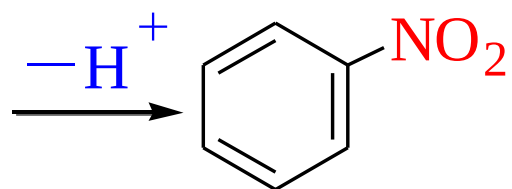
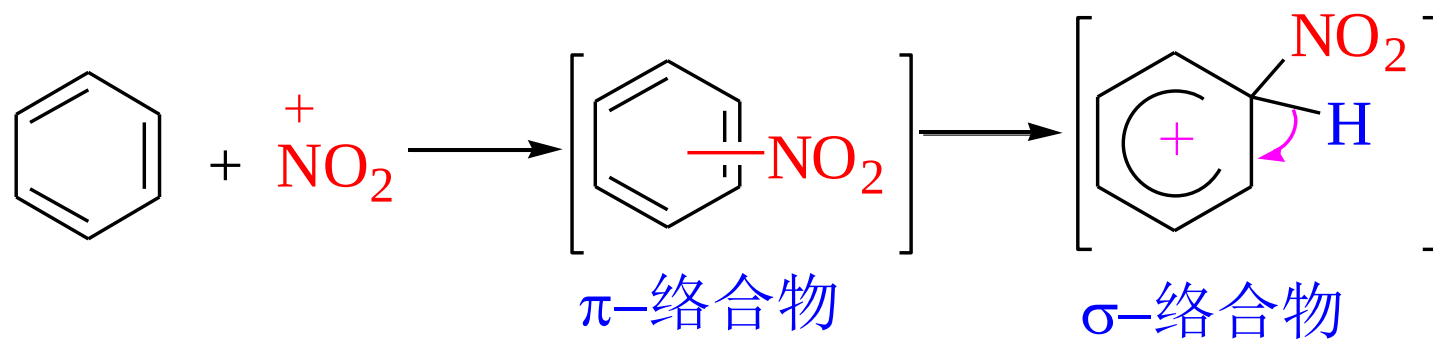
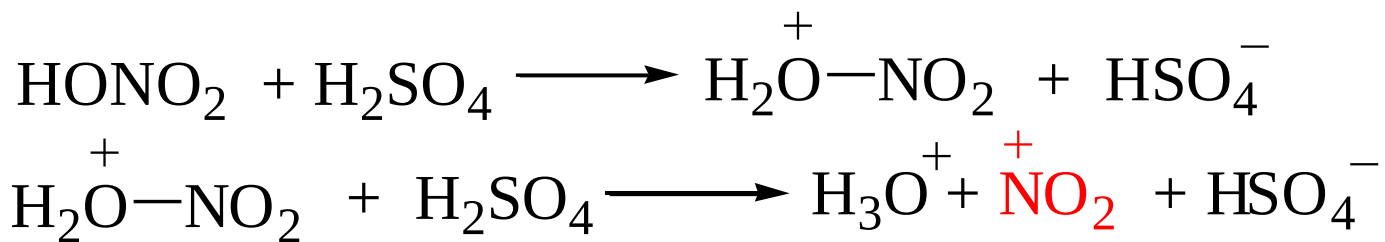
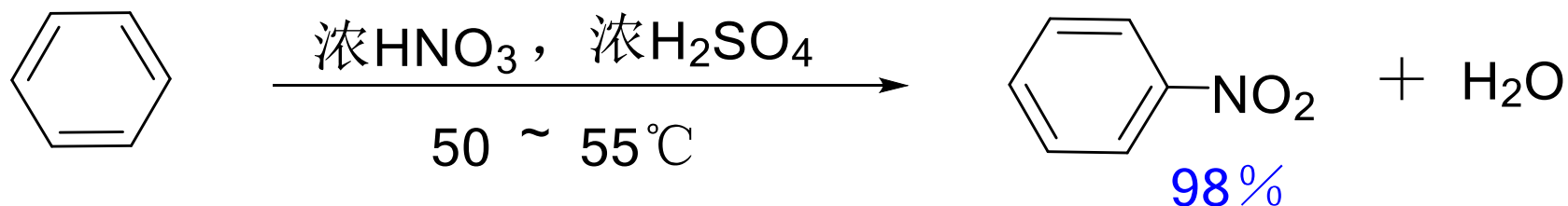
# 芳香亲电取代反应

(Electrophilic Aromatic Substitution)

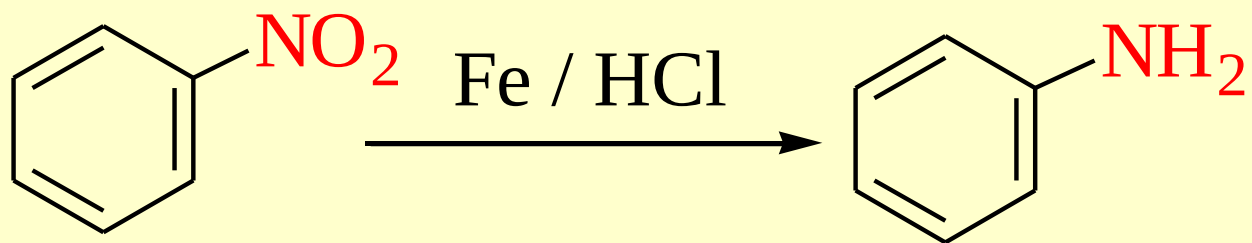


$\pi$ 络合物

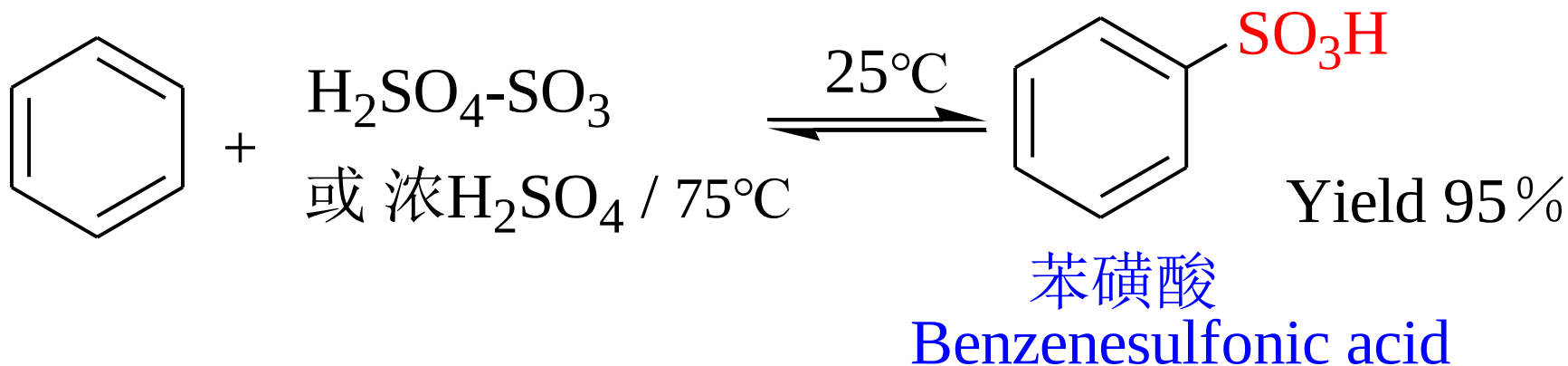
碳正离子  
( $\sigma$ 络合物)



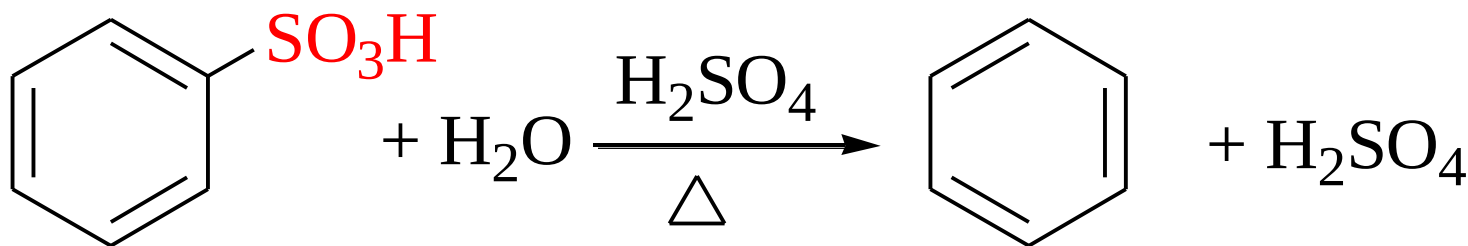
# 硝基苯还原可得苯胺



# 磺化反应



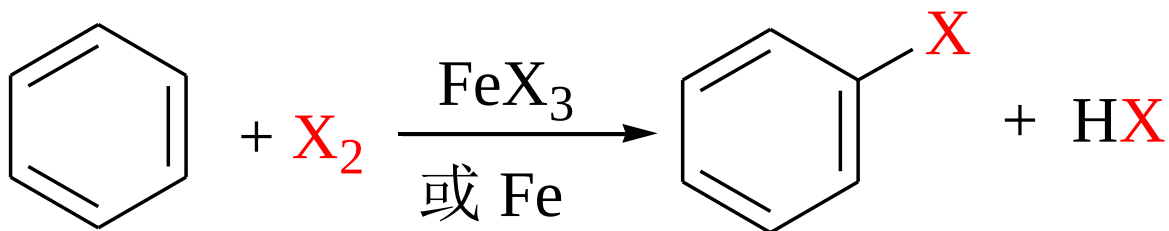
- 苯磺酸在稀酸溶液中加热，磺酸基水解脱落



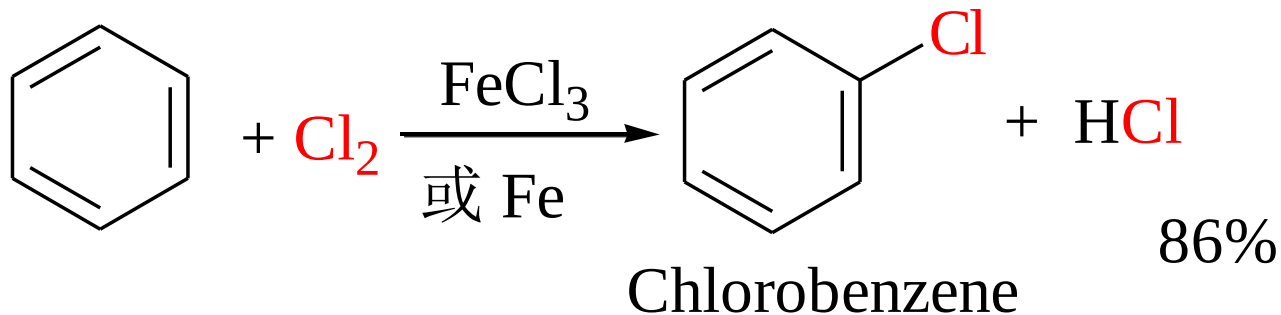
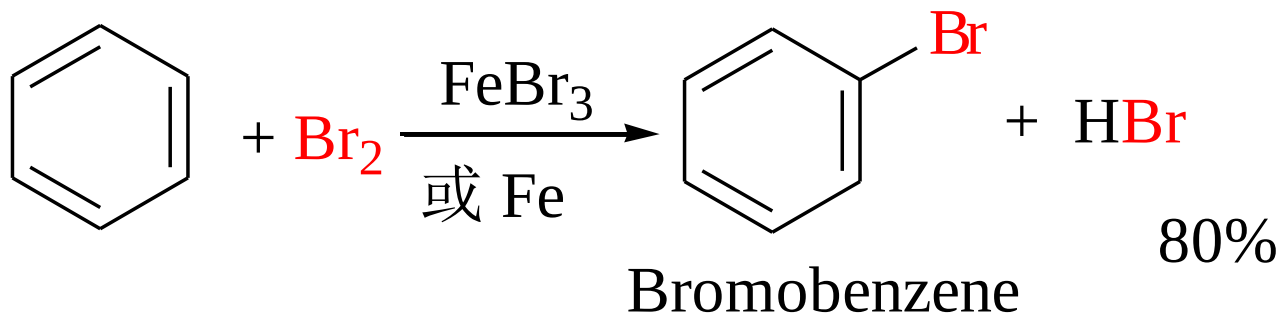
磺化反应——占位效应具有重要用途



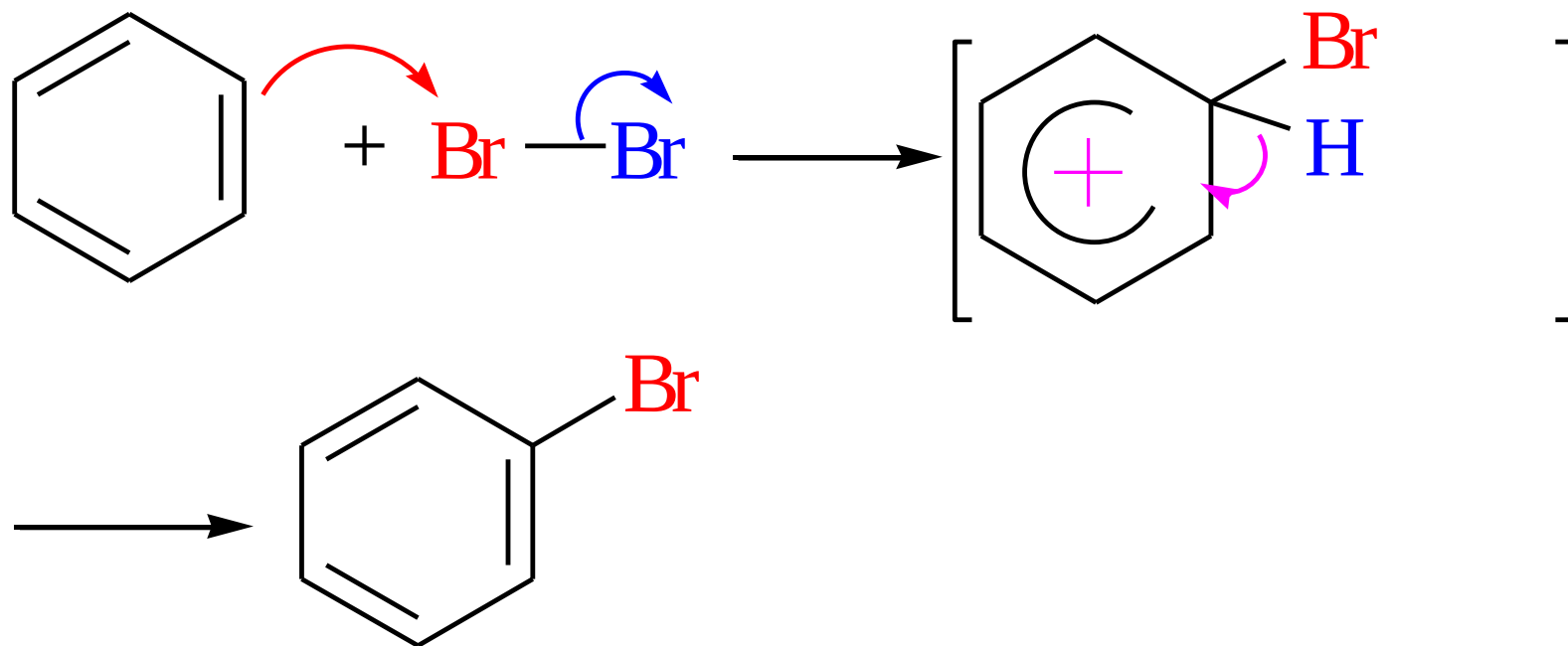
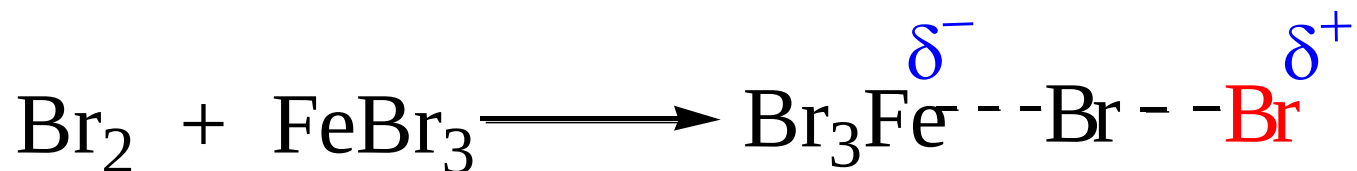
# 卤化反应



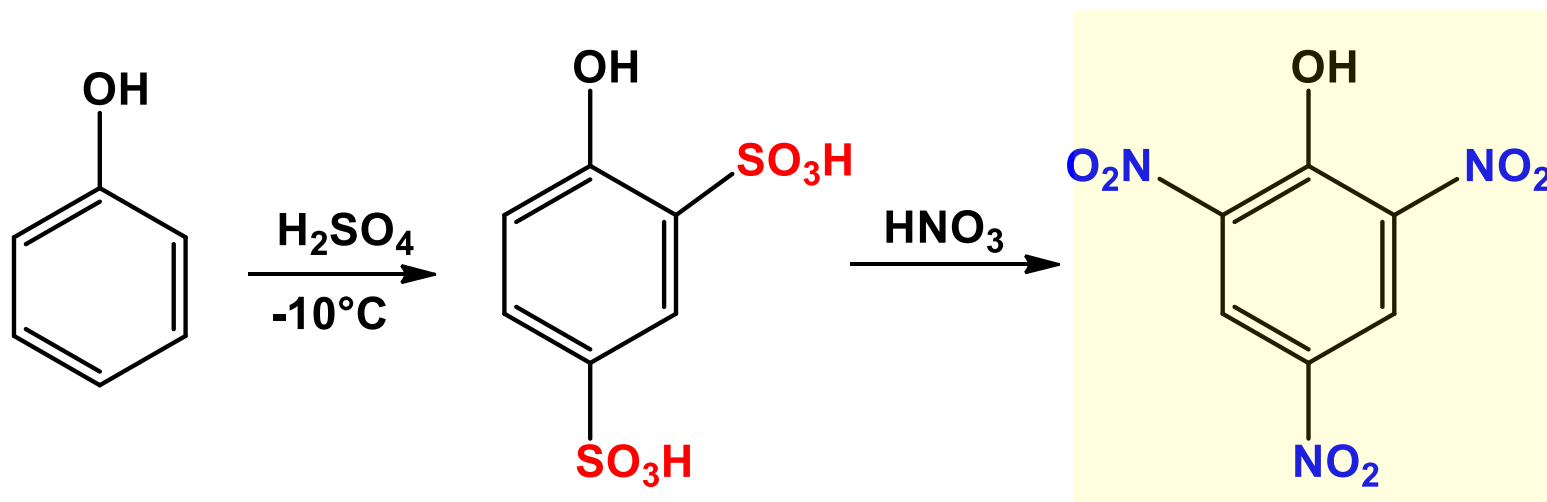
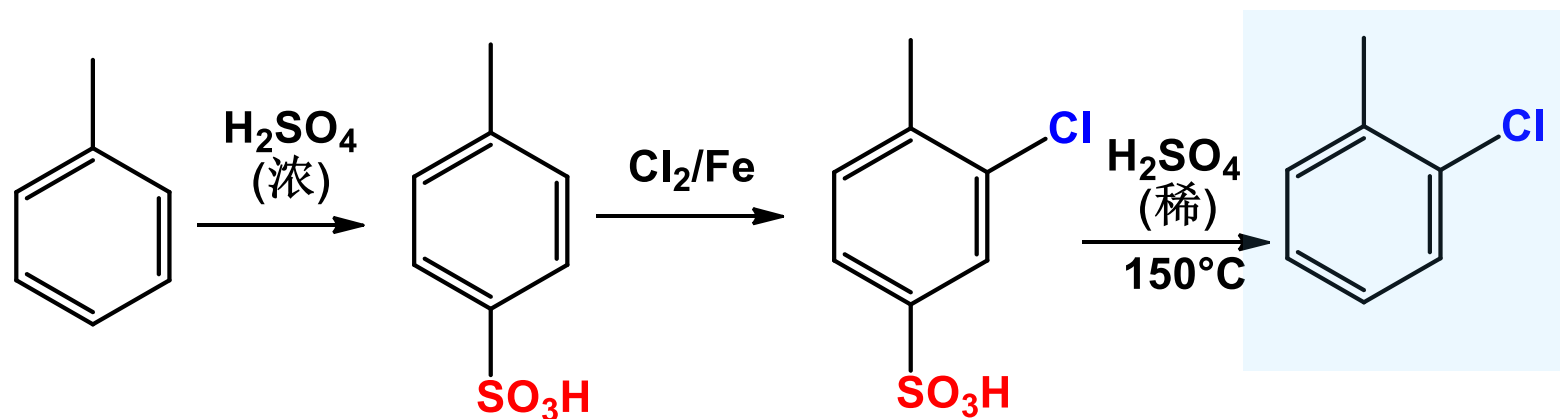
反应活性:  $F_2 > Cl_2 > Br_2 > I_2$



# 反应机理



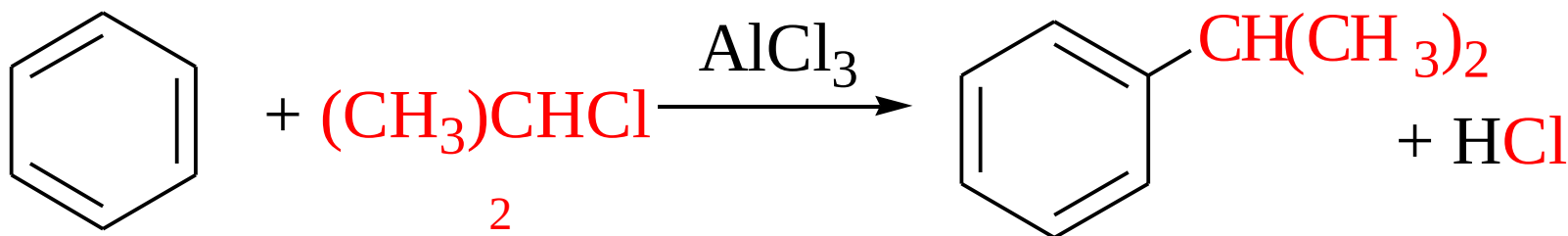
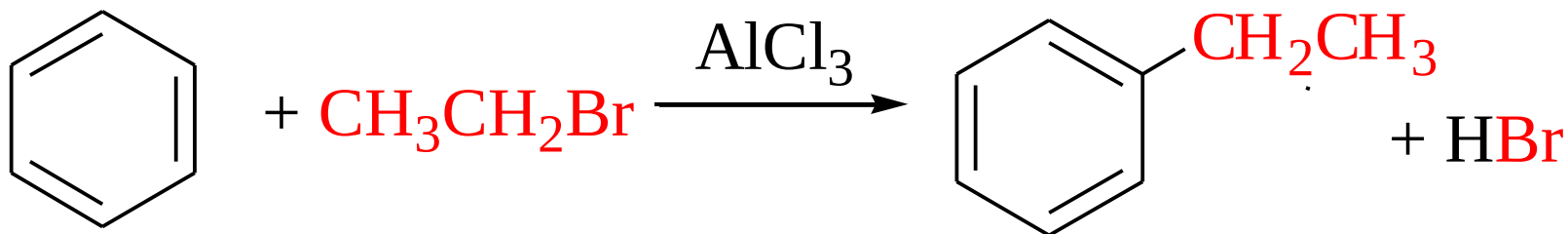
## 利用磺化反应的可逆性，保护芳环某些特定位置



苦味酸

# 傅—克（Friedel—Crafts）反应

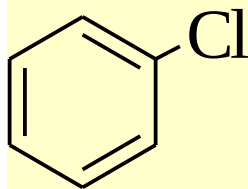
## (A) 烷基化(Alkylation)



催化剂 Lewis 酸

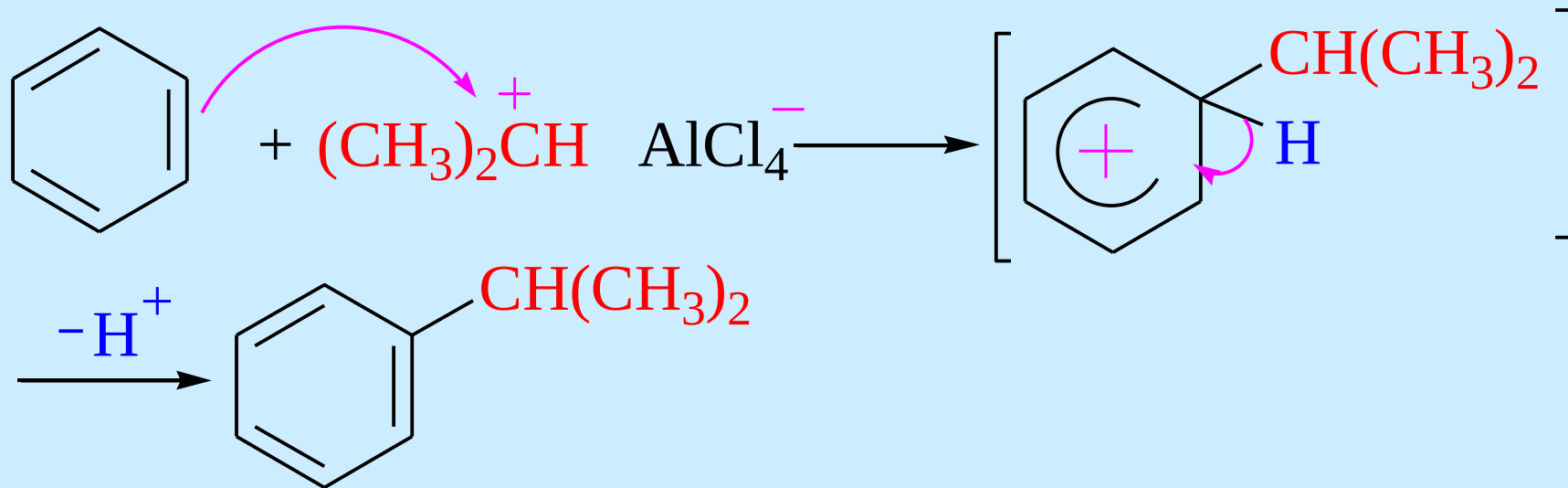
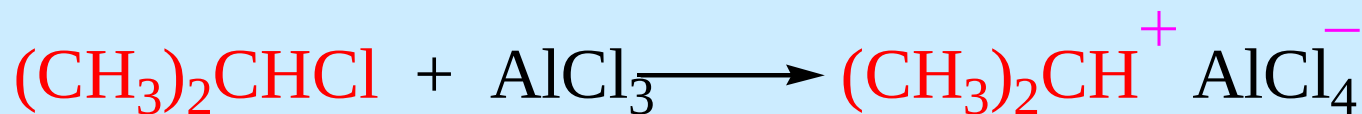
$\text{FeCl}_3$ ,  $\text{SnCl}_4$ ,  $\text{ZnCl}_2$

$\text{HF}$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{H}_3\text{PO}_4$ ,  $\text{BF}_3$

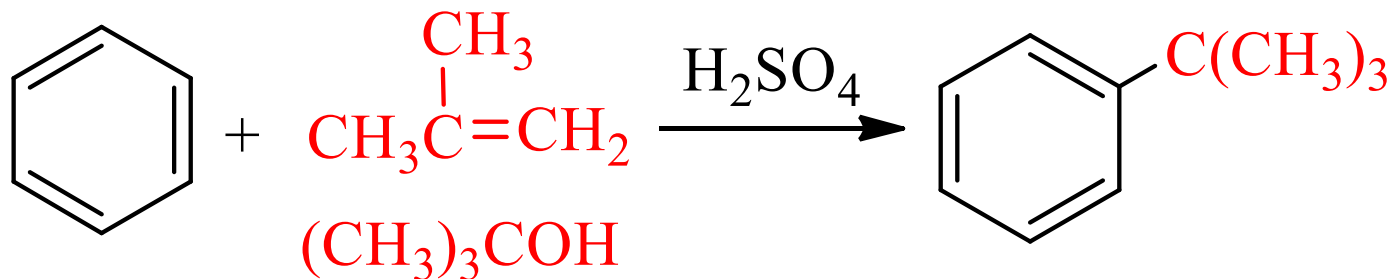
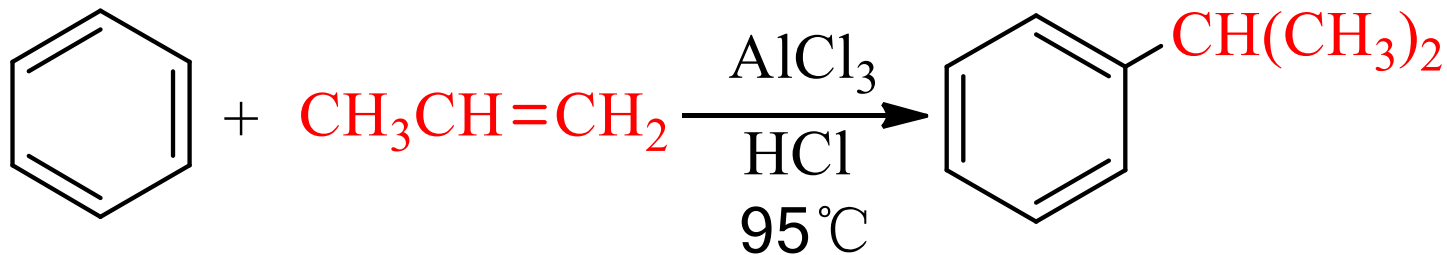
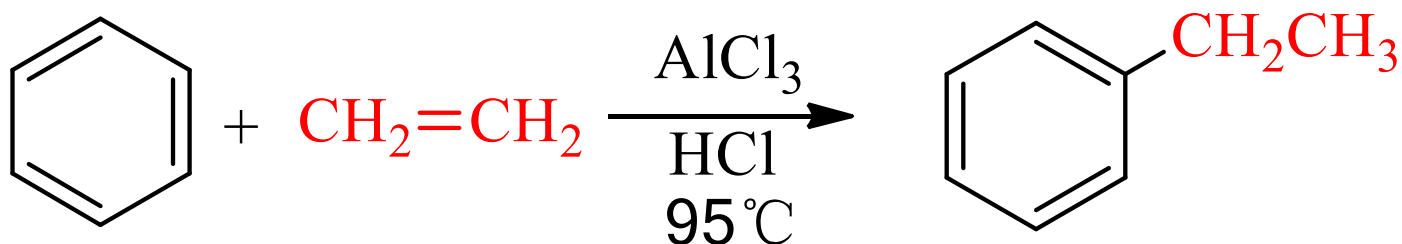
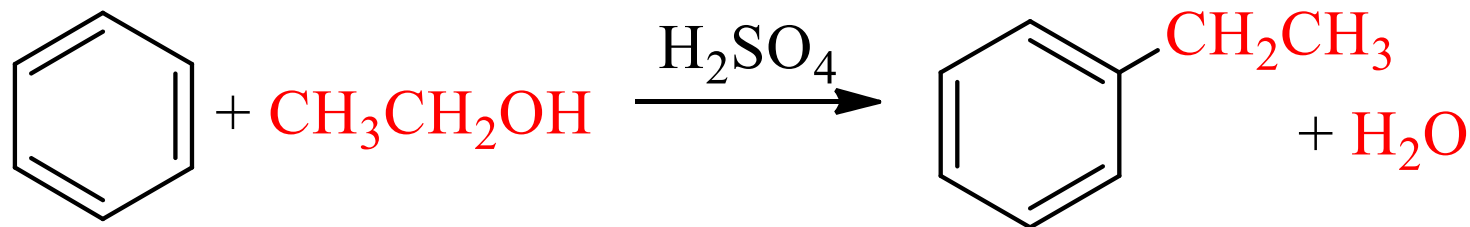


$\text{CH}_2=\text{CHCl}$  不反应

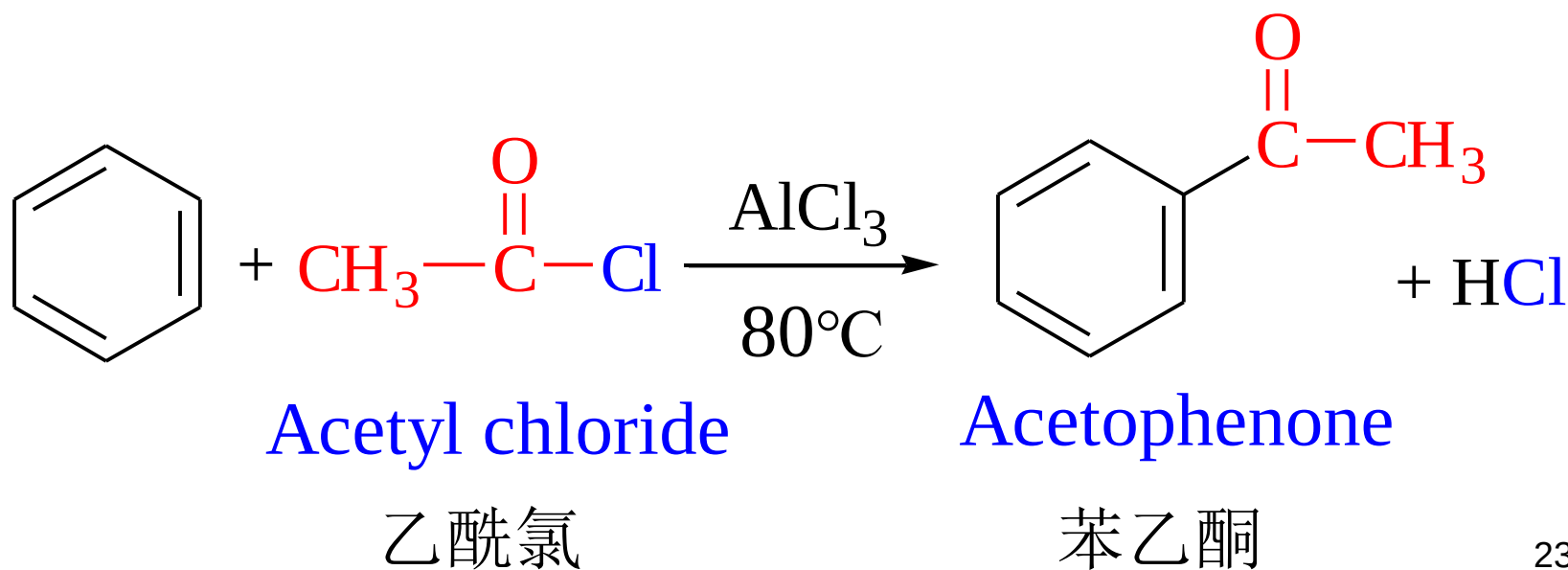
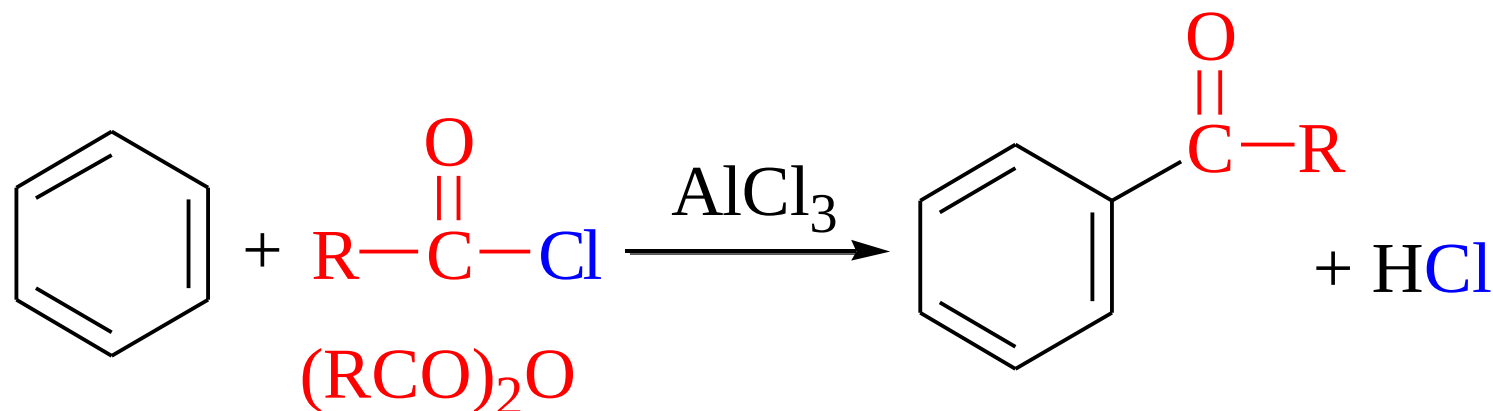
# 反应机理



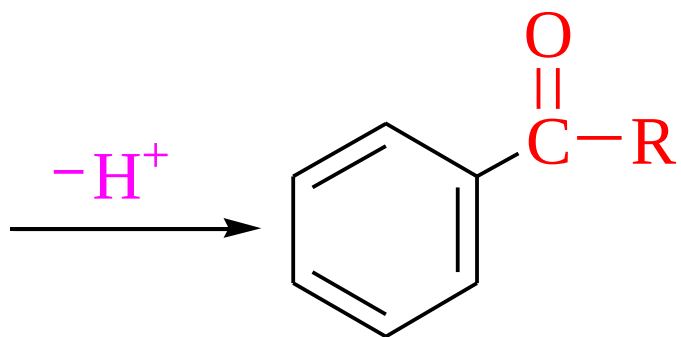
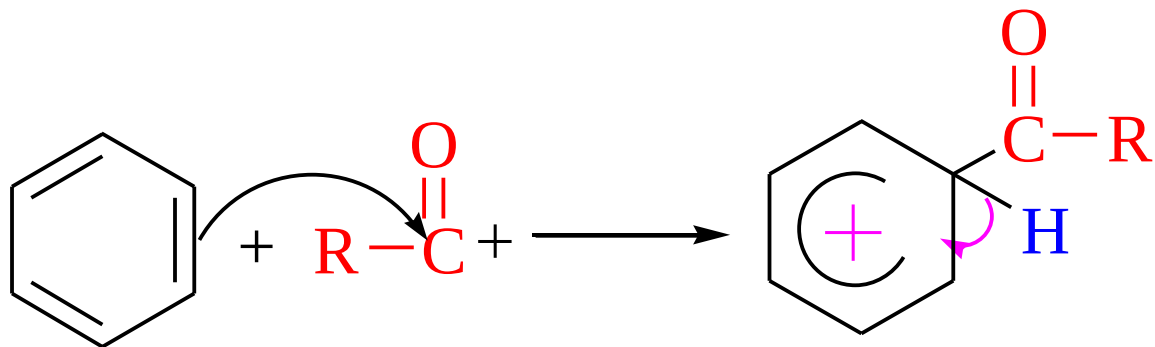
烷基化试剂除卤代烃外，还可以用醇、烯烃



## (B) 酰基化(Acylation)

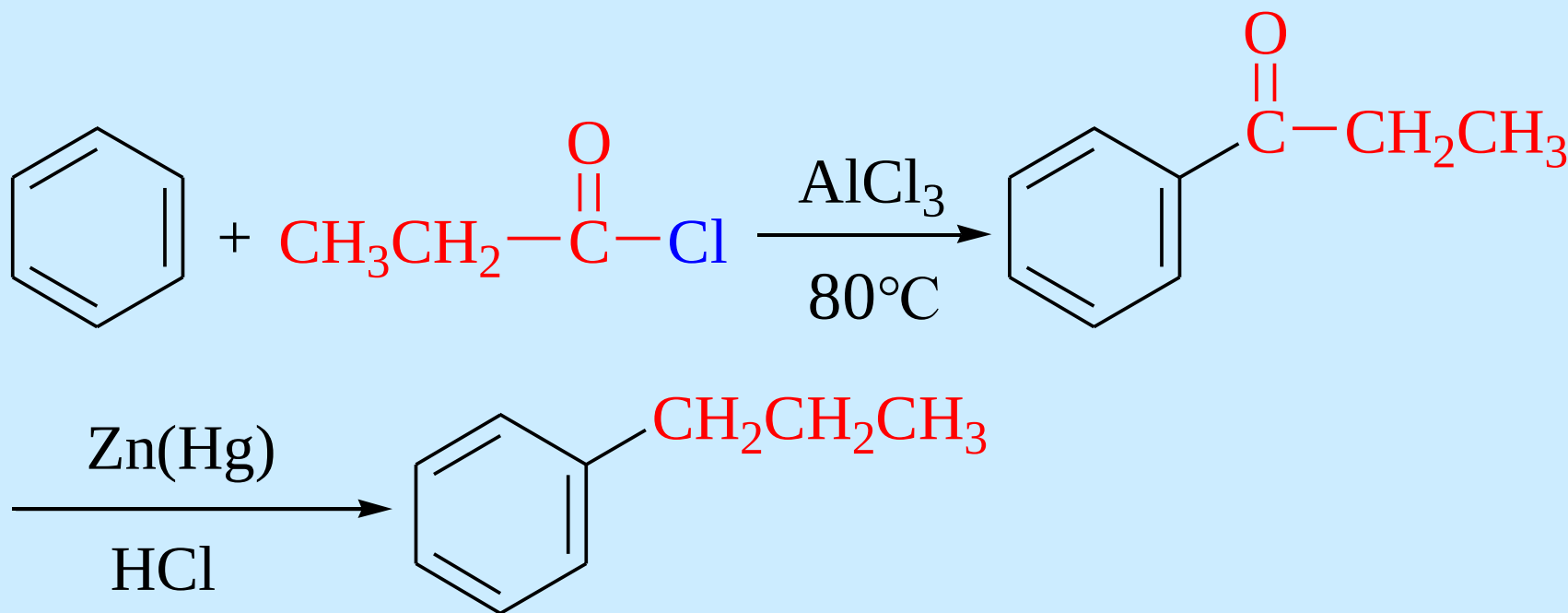


# 反应机理



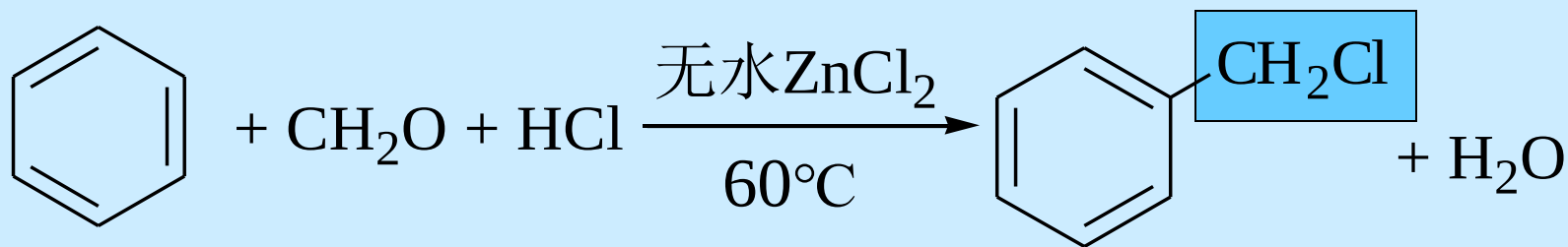


# 酰基化—还原反应系列

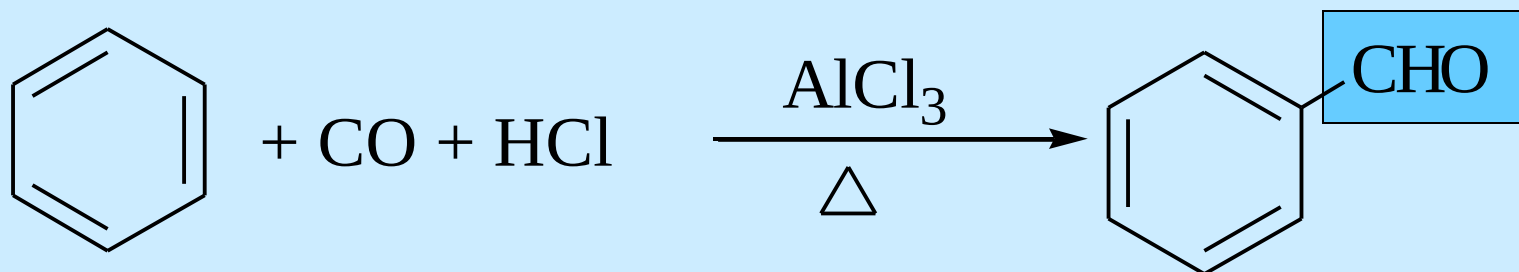


**Clemmenson reduction**

克莱门森还原



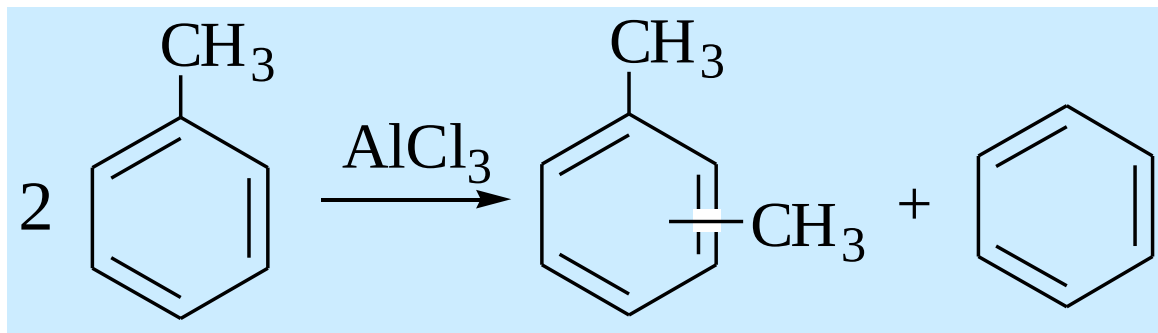
## 氯甲基化 (Chloromethylation)



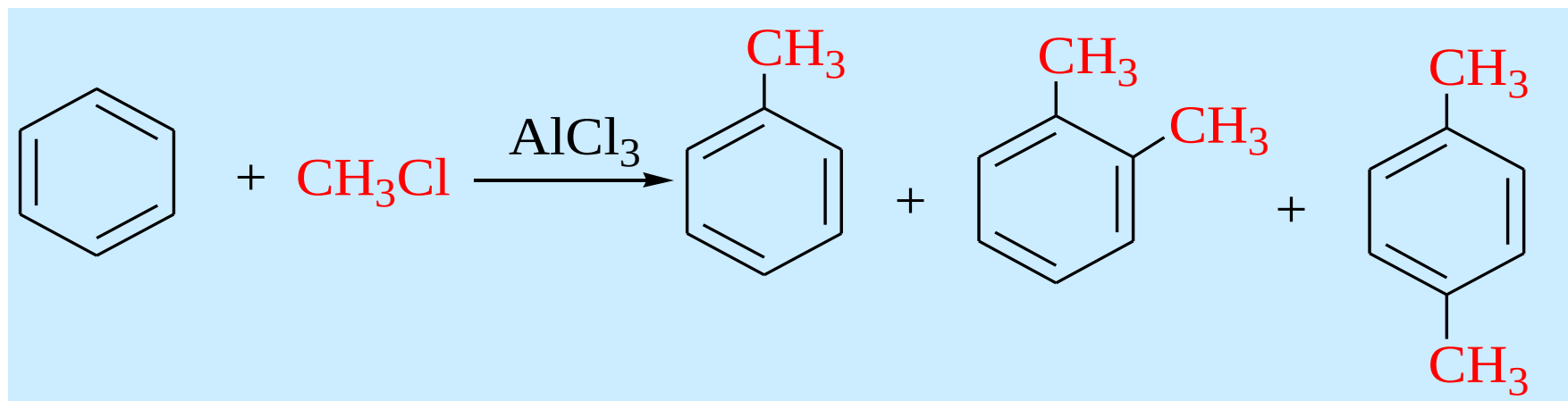
## Gatterman—Koch反应

# Friedel—Crafts反应局限性

- 烷基化是可逆反应

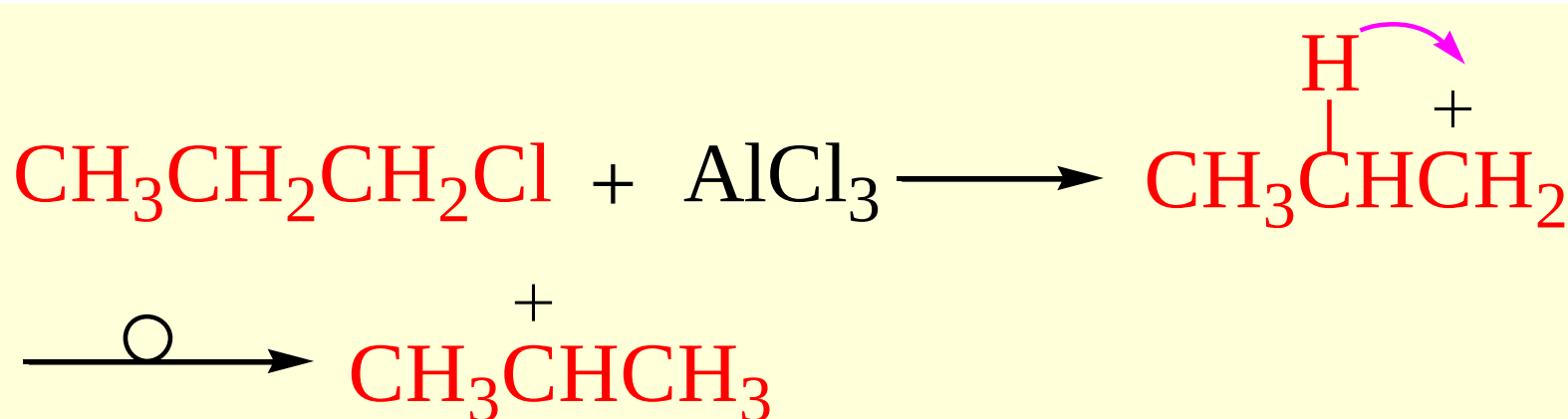
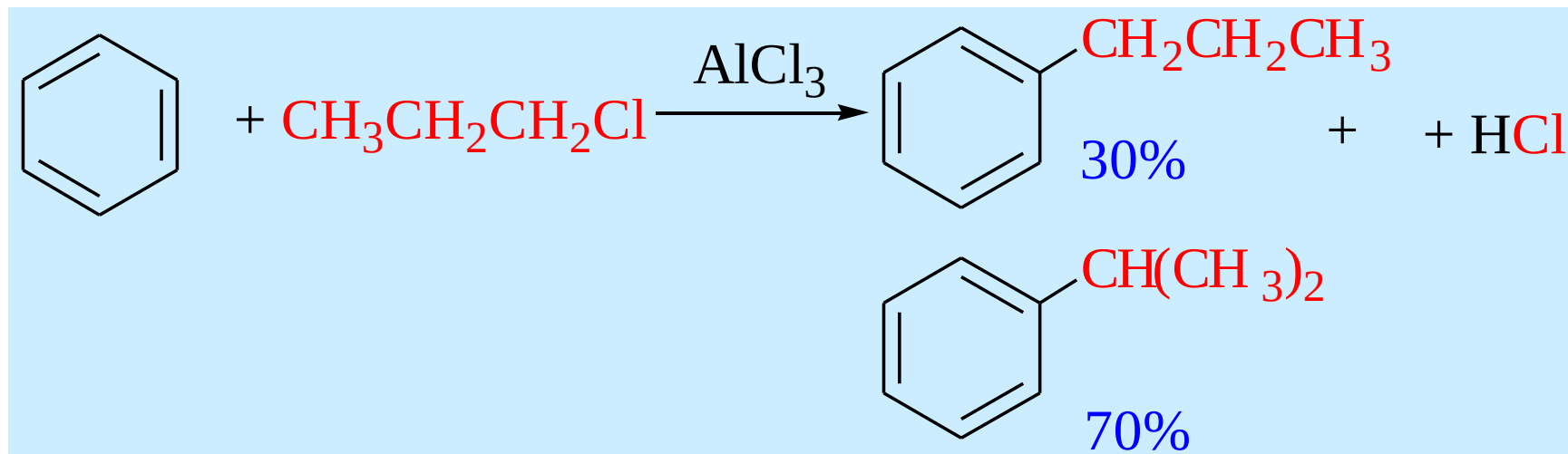


- 易发生多取代现象



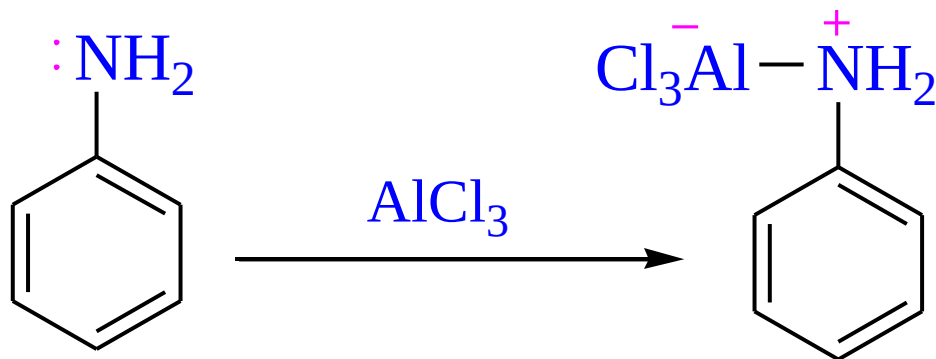
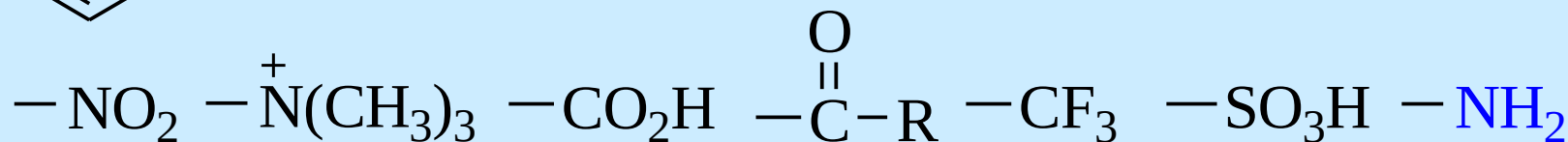
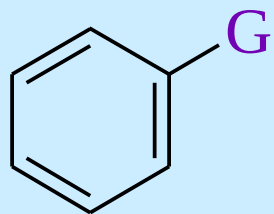
# Friedel—Crafts反应局限性

- 当RX超过3个碳原子时，可能发生重排反应



# Friedel—Crafts反应局限性

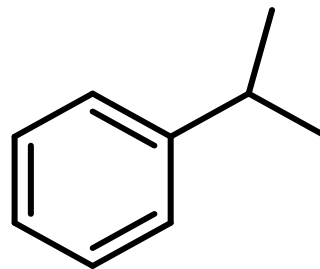
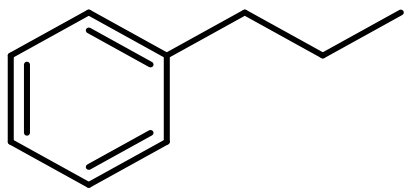
- 当苯环上有强吸电子取代基时，不发生Friedel—Crafts反应



不发生  
Friedel-Crafts 反应

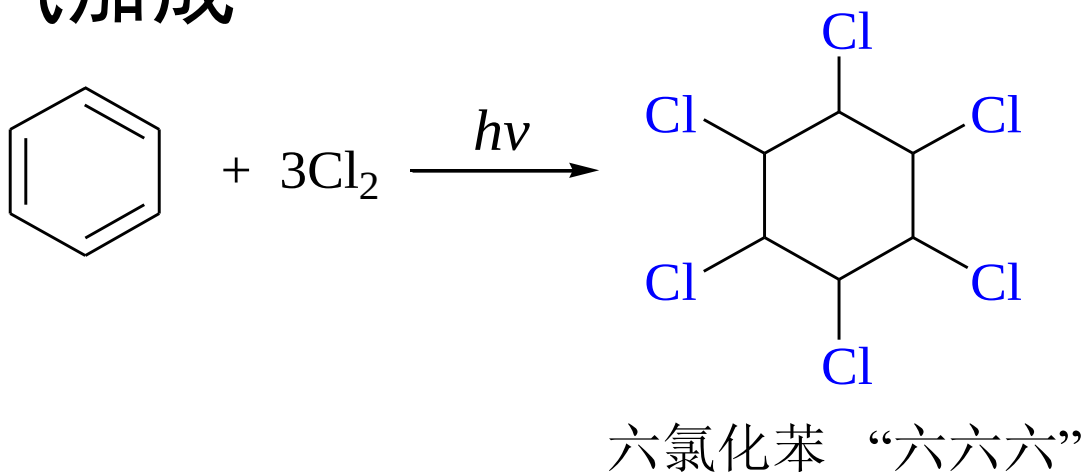
# 测试

合成下面化合物

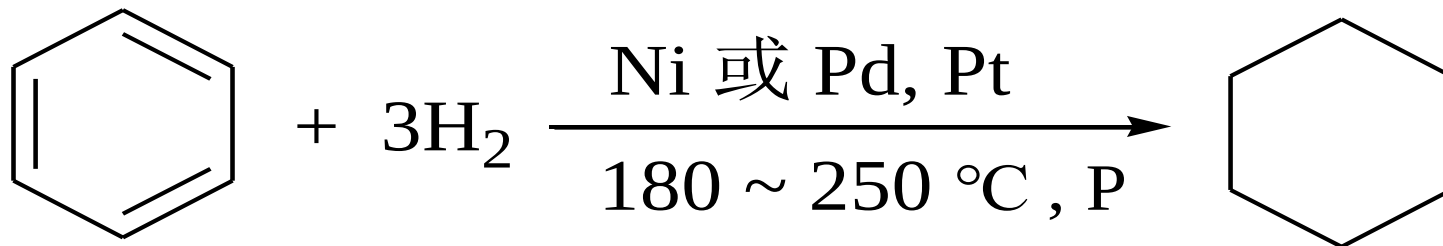


# 加成与还原反应

- 与氯气加成

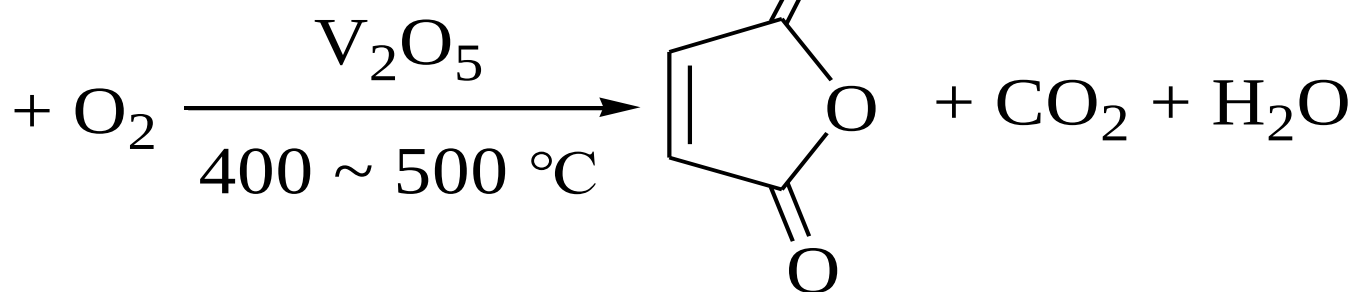
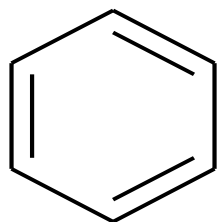
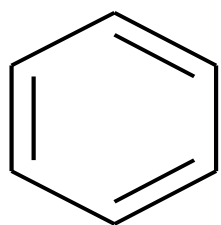


- 加氢



# 氧化反应

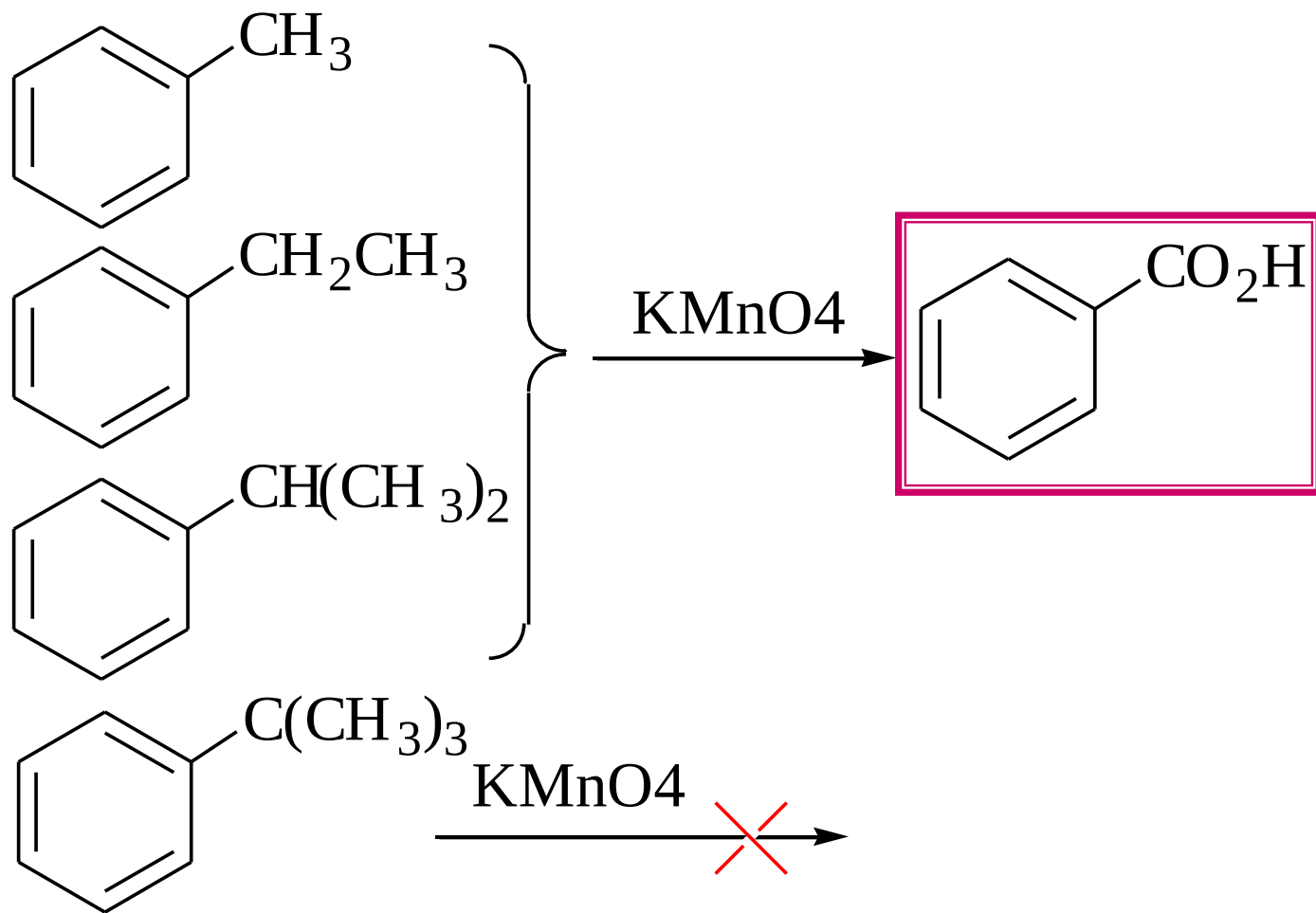
- 苯环的氧化

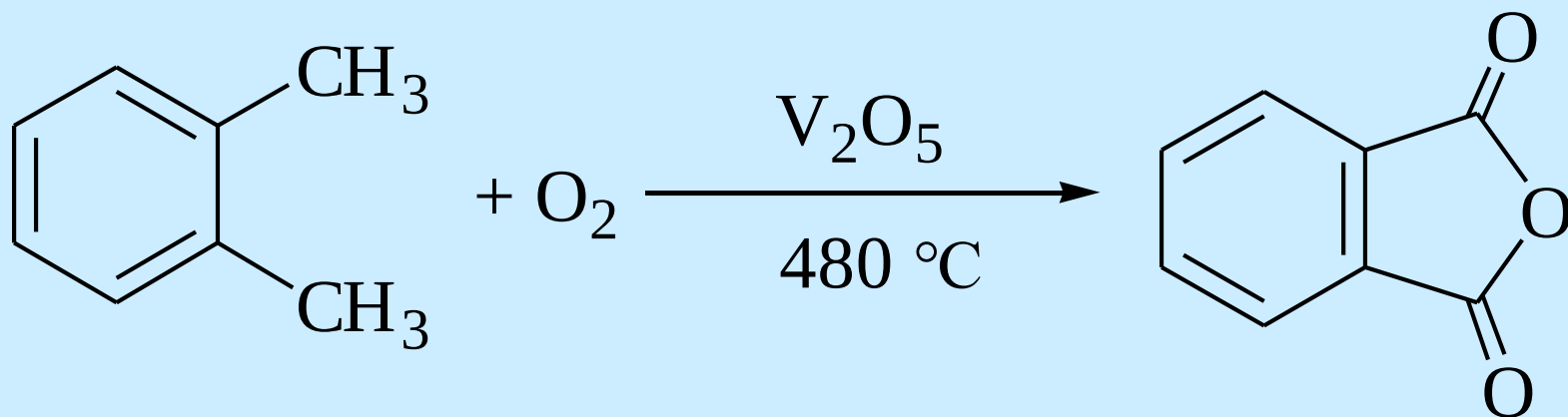
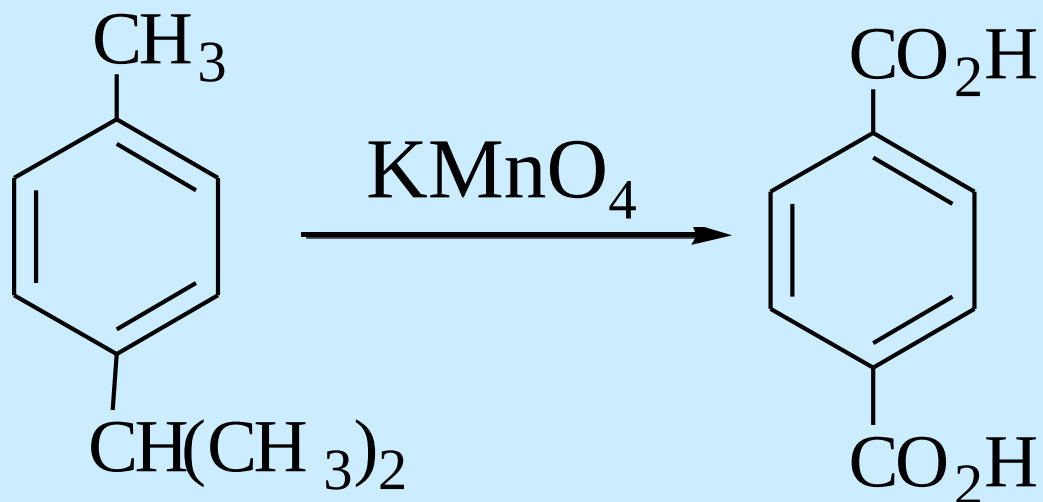


顺丁烯二酸酐  
Maleic anhydride



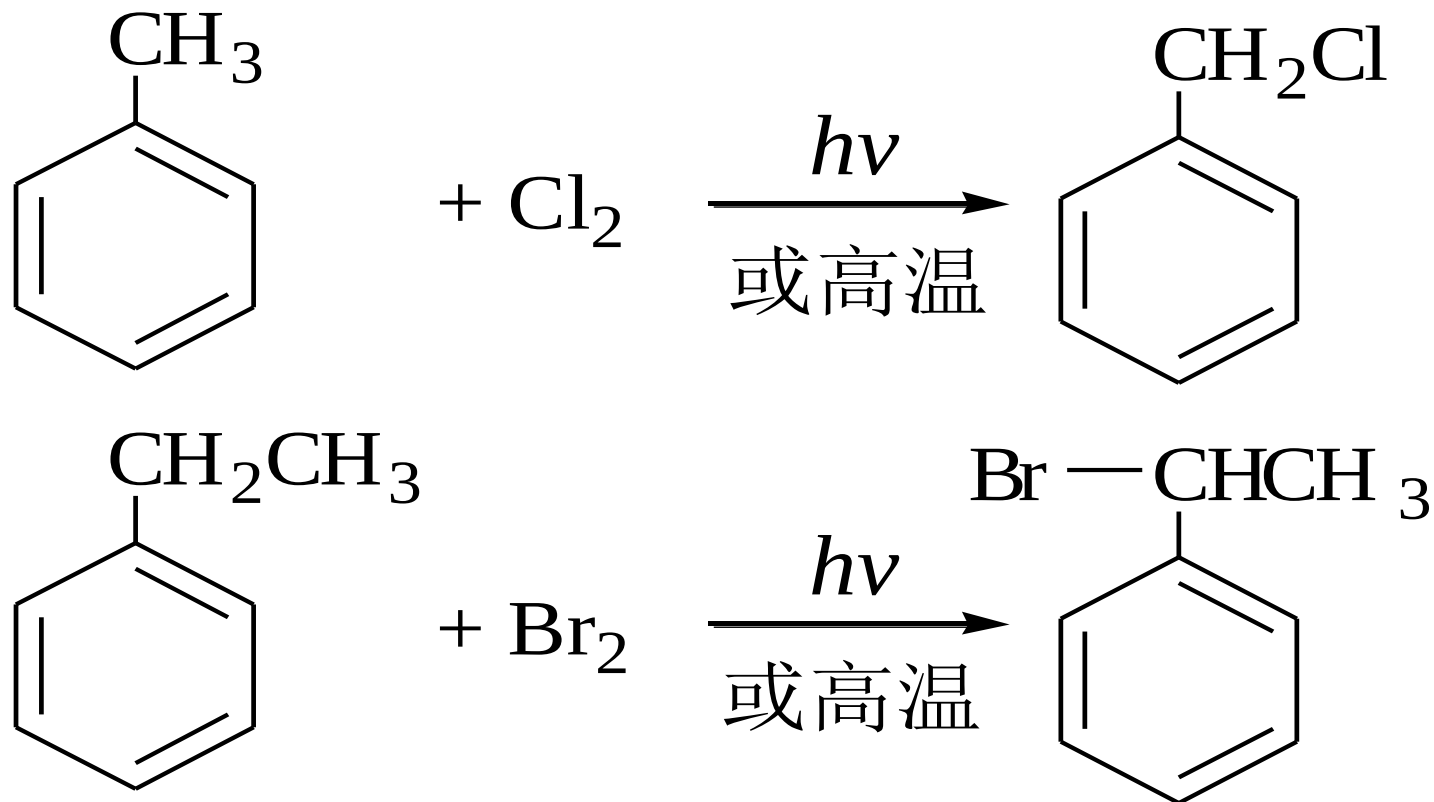
## • 侧链氧化





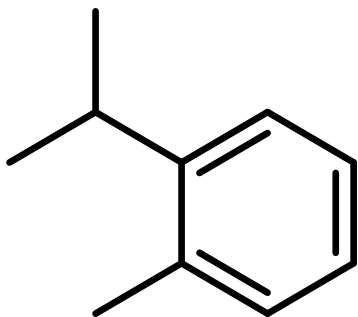
邻苯二甲酸酐

# $\alpha$ -卤代反应



# 测试

写出下列产物



KMnO<sub>4</sub>

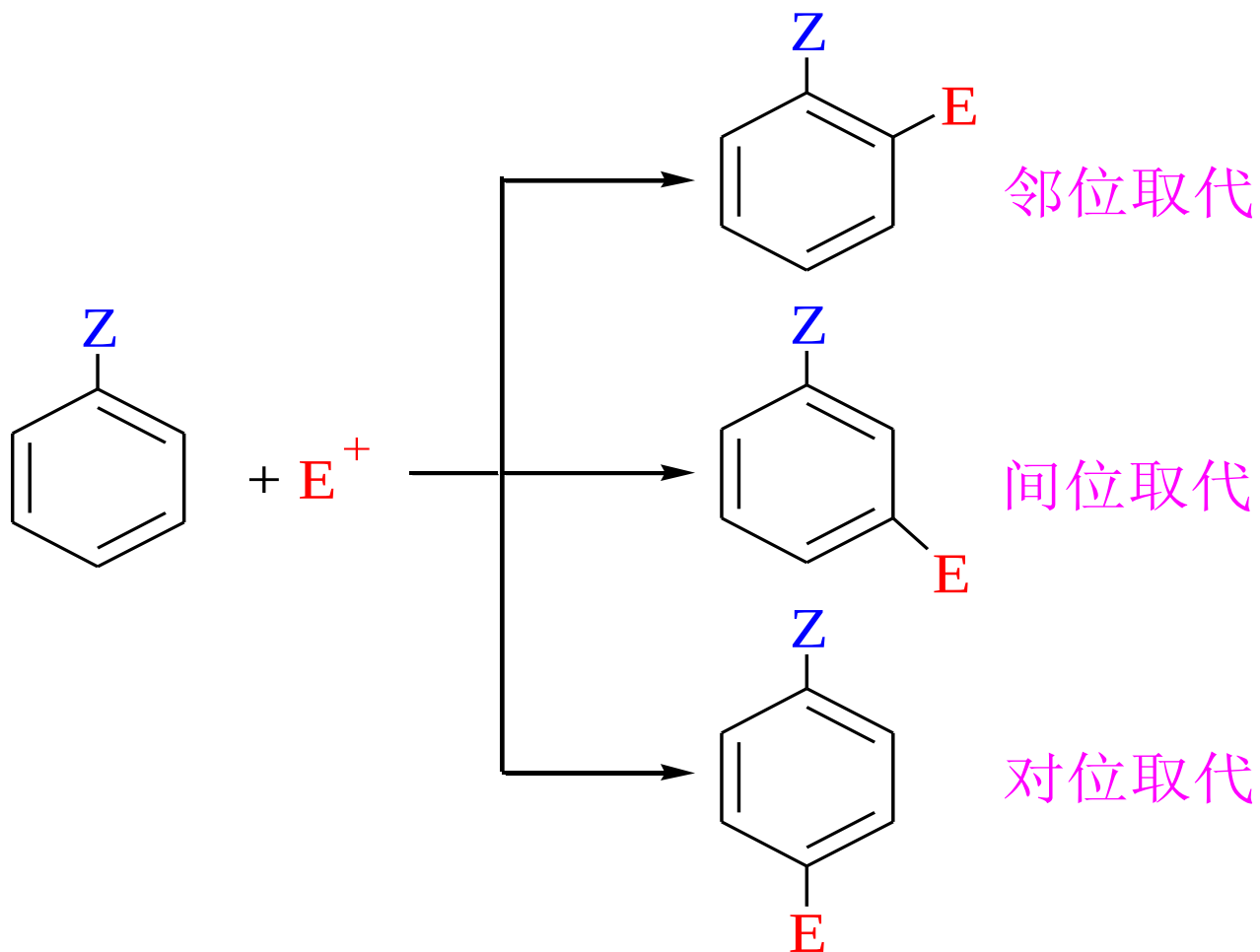


高温

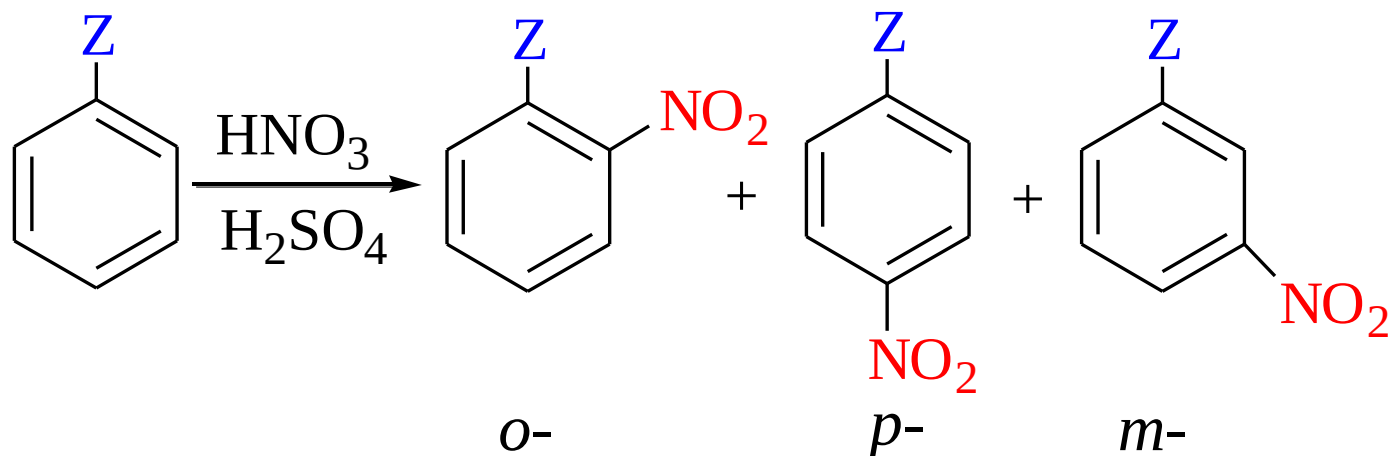


# 苯环上亲电取代的定位规律

## 定位效应(Orientation)



# 定位效应

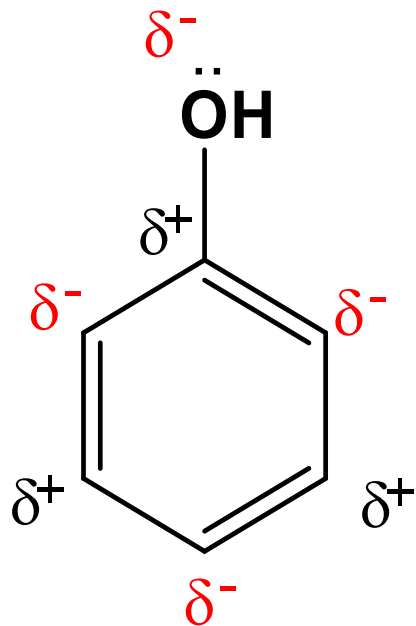


**Z =      OH      Me      Cl      NO<sub>2</sub>      CN**

|      |    |    |    |    |    |
|------|----|----|----|----|----|
| o(%) | 40 | 58 | 30 | 6  | 17 |
| p(%) | 60 | 38 | 69 | <1 | 2  |
| m(%) | <1 | 4  | 1  | 93 | 81 |

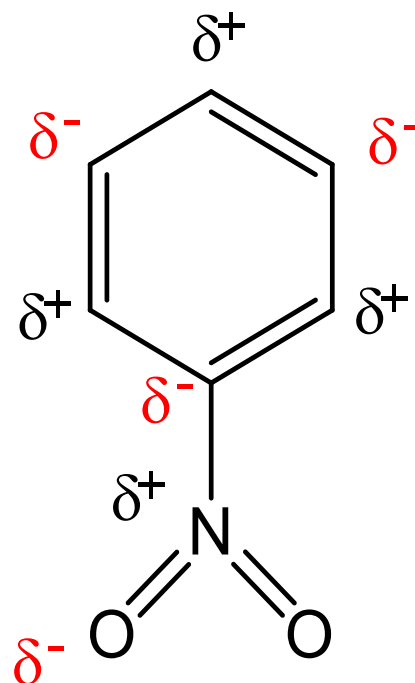
**定位效应**：取代基进入的位置受原有取代基的影响

# 定位效应



**-OH**

邻对位定位基

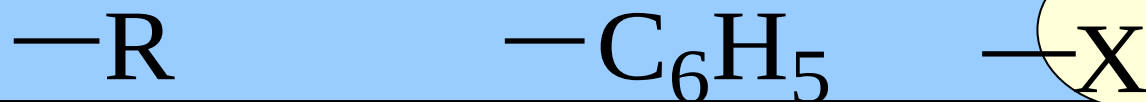
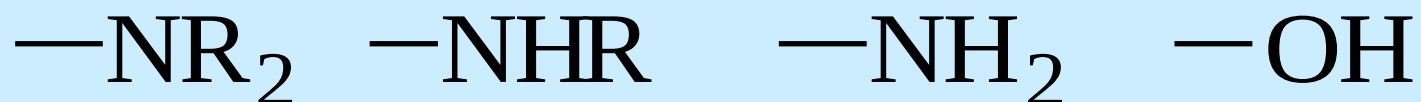


**-NO<sub>2</sub>**

间位定位基

本质：外来亲电试剂只能进攻碳  $\delta^-$

## 第一类定位基(邻对位定位基)



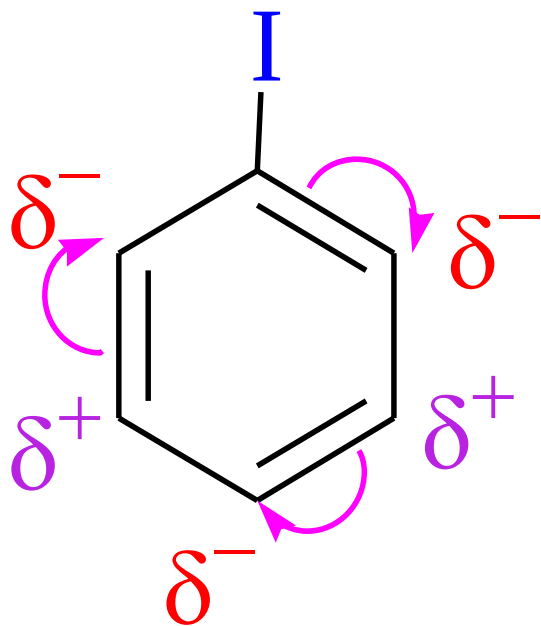
## 第二类定位基(间位定位基)



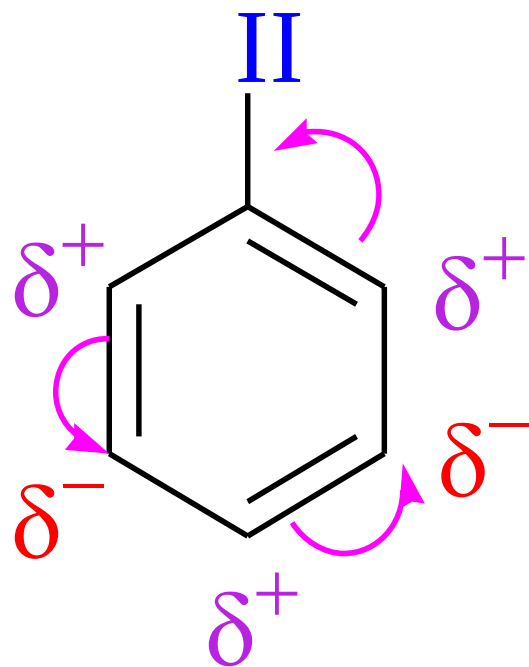


# 定位规律的理论解释

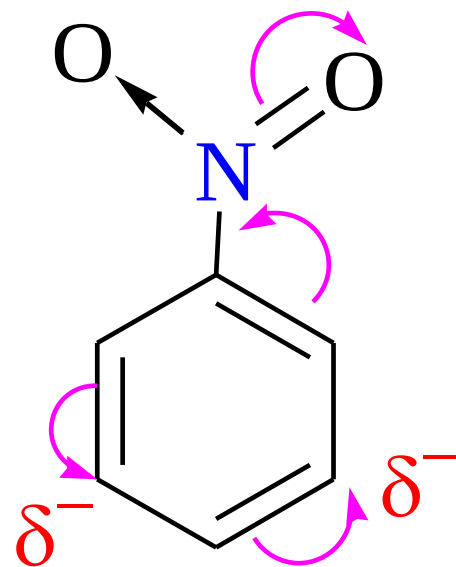
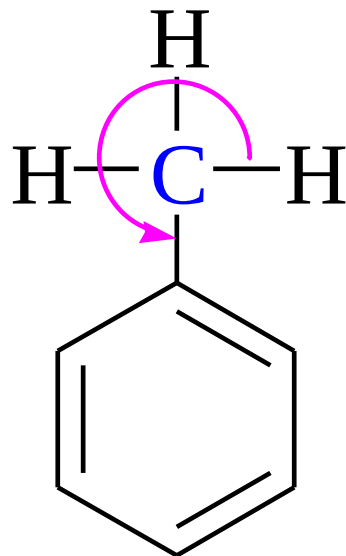
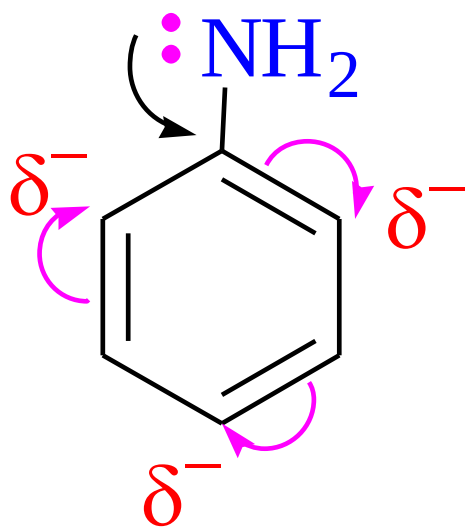
## 苯环上的电子分布情况



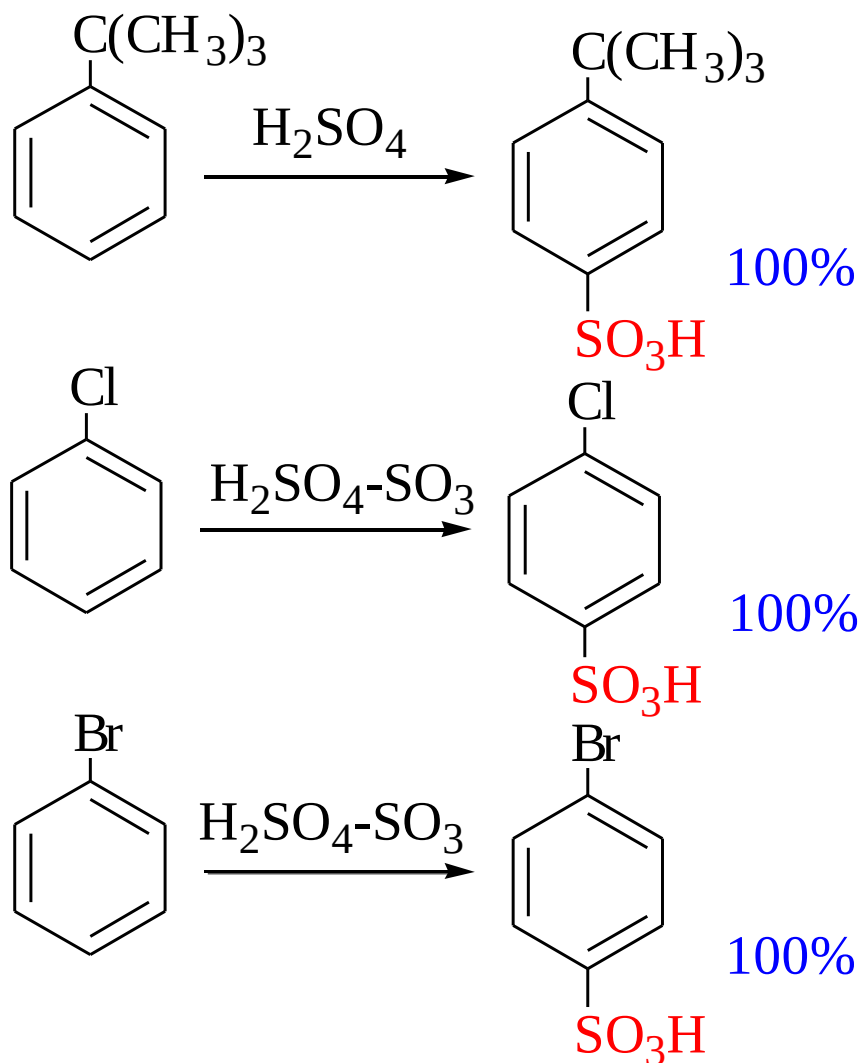
I 类定位基



II 类定位基

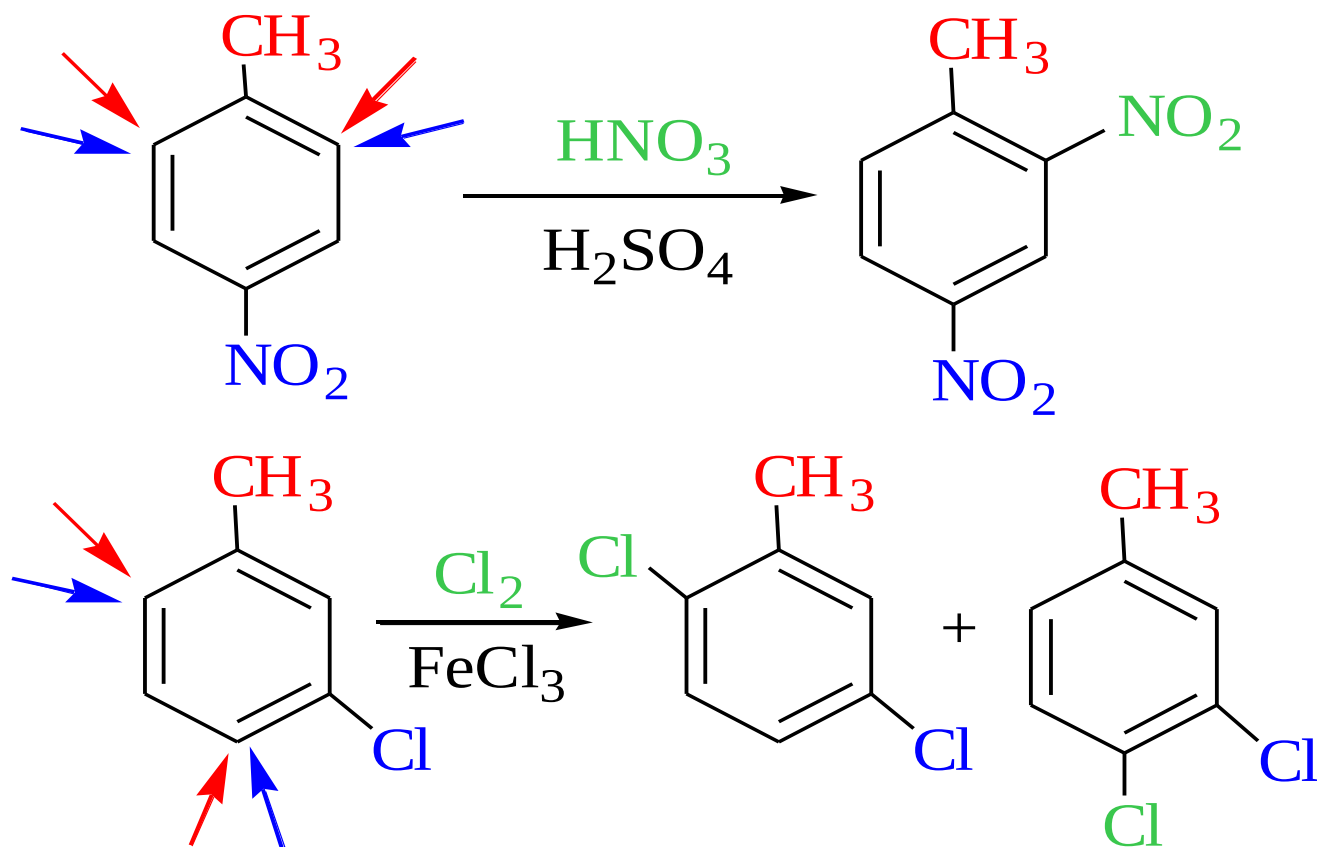


# 取代基与试剂体积都大时，邻位产物极少



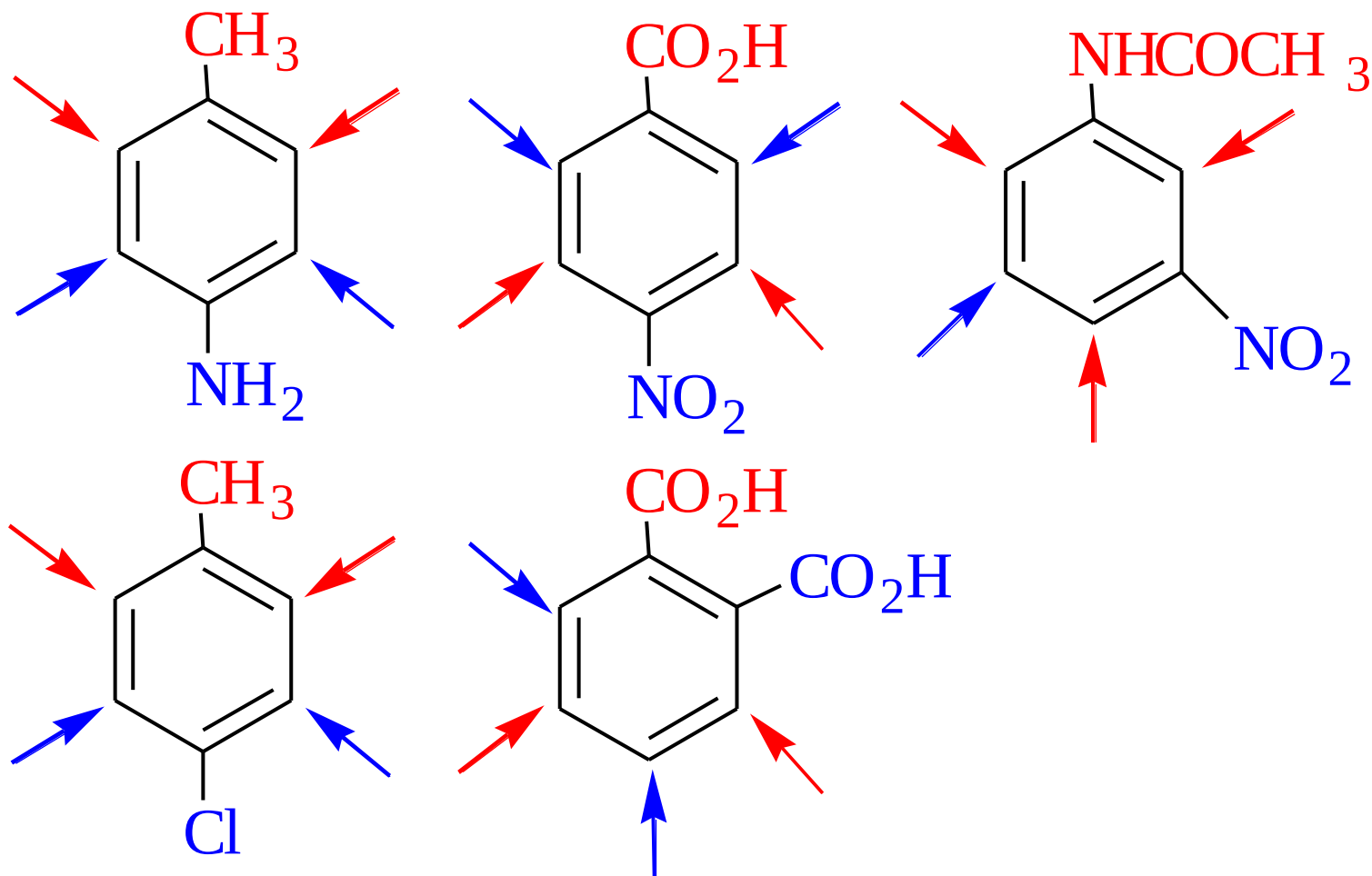
# 二元取代苯的定位

## 1. 两个取代基定位效应一致

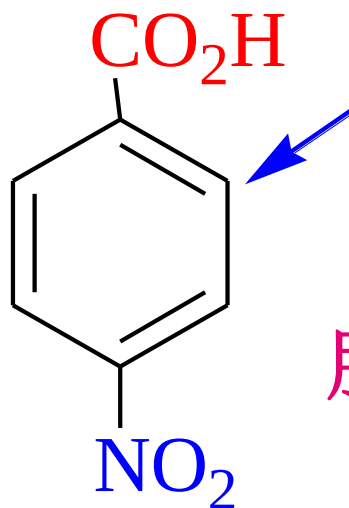
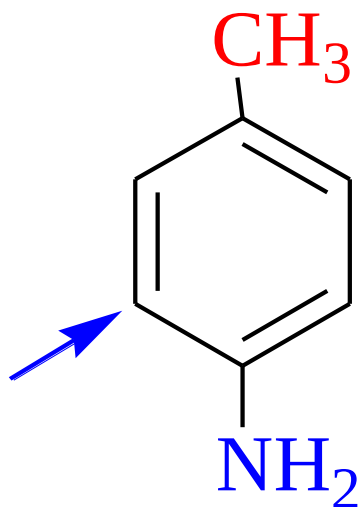


# 二元取代苯的定位

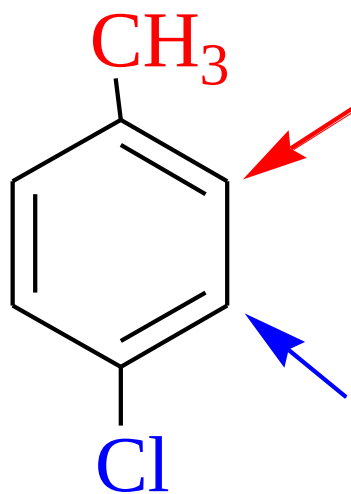
## 2. 两个取代基定位效应不一致



## 两个取代基同类

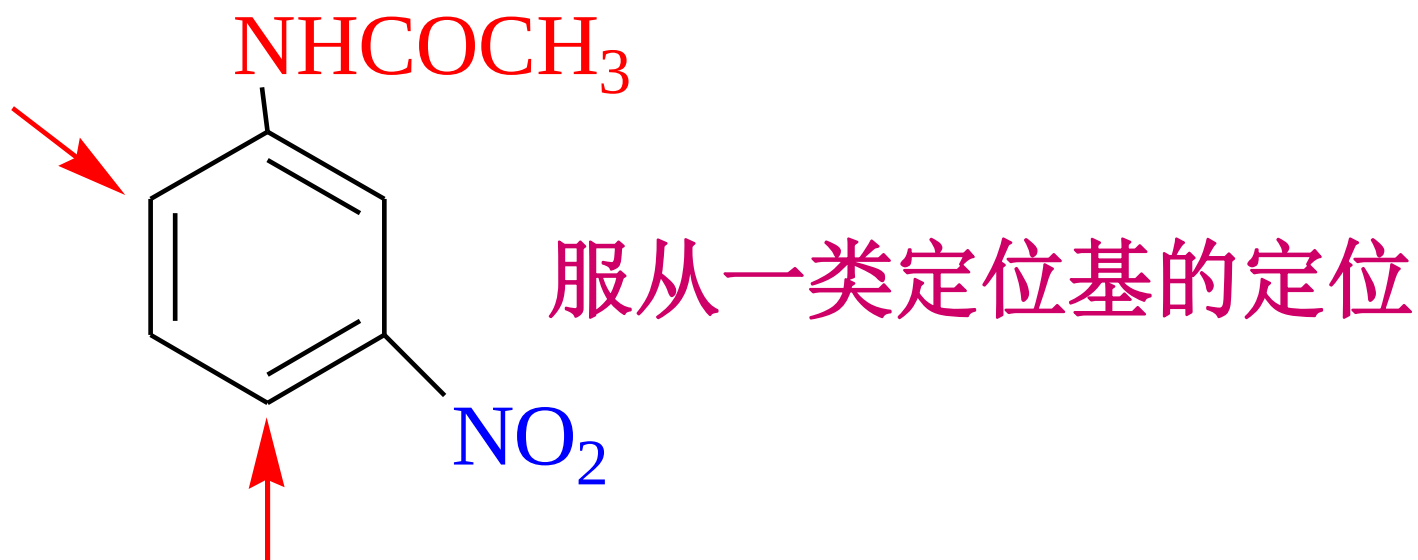


服从较强取代基的定位

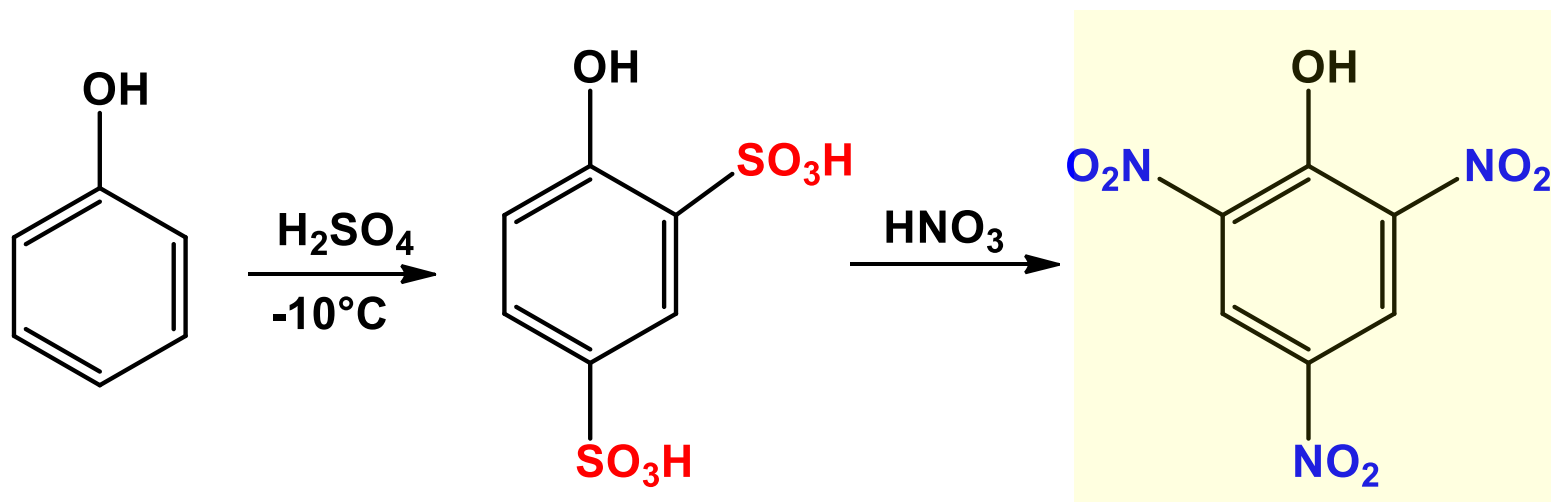
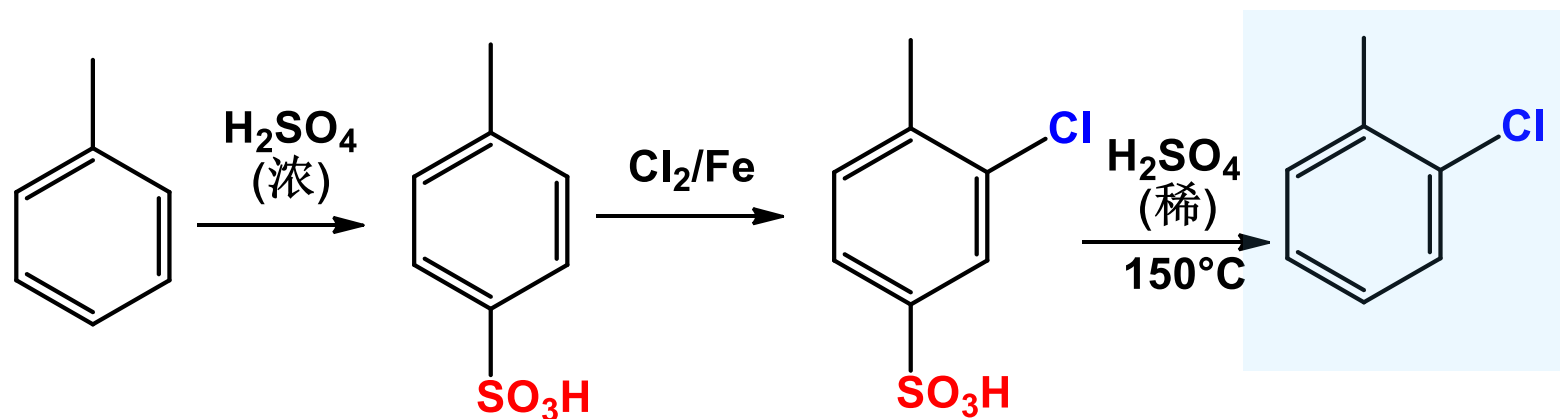


定位能力差不多时  
两者都有

## 两个取代基不同类



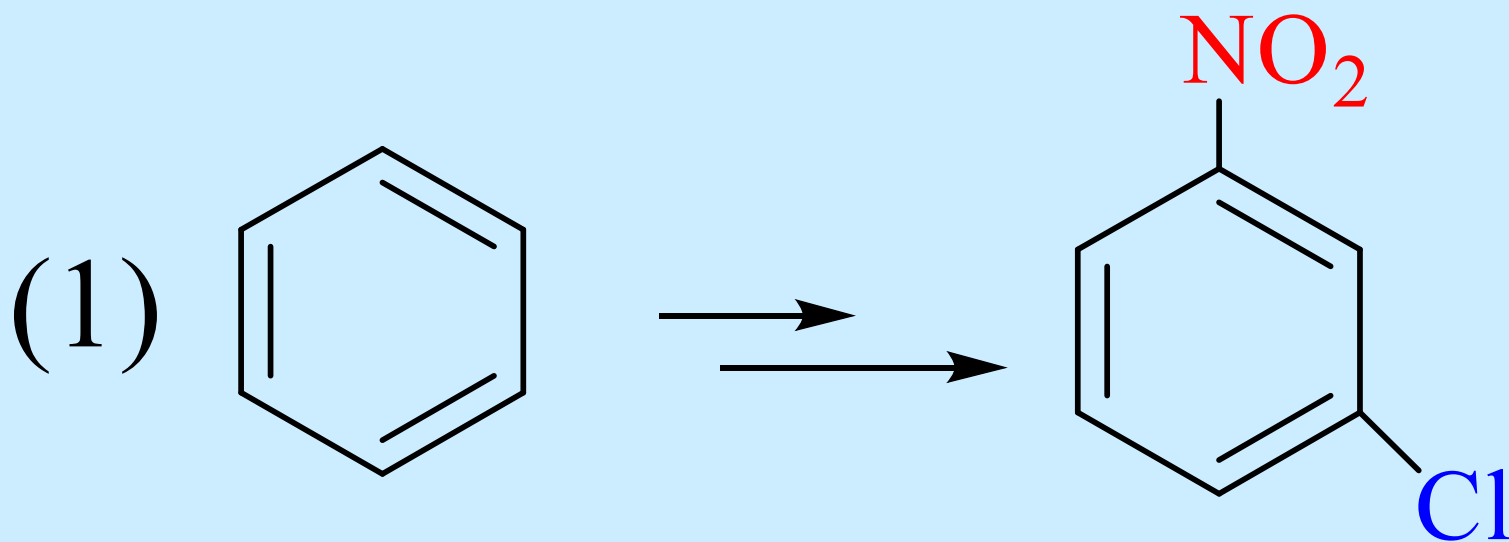
## 利用磺化反应的可逆性，保护芳环某些特定位置



苦味酸



# 定位规律的应用

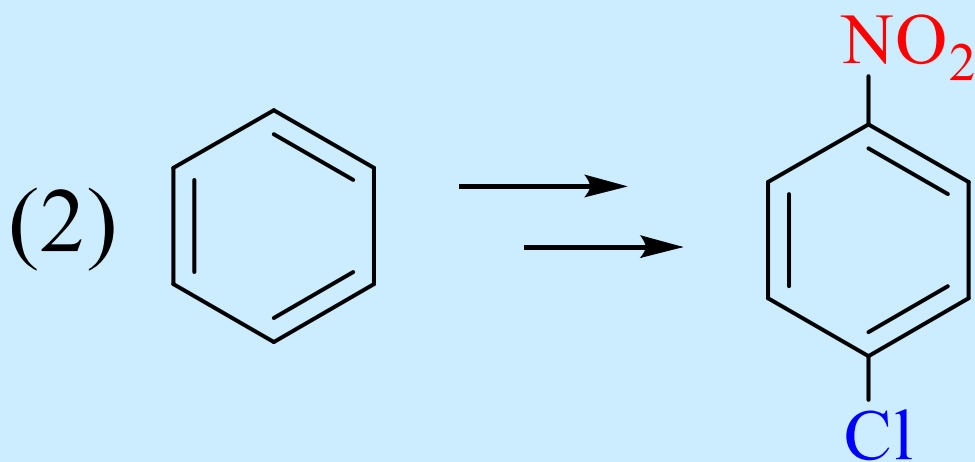
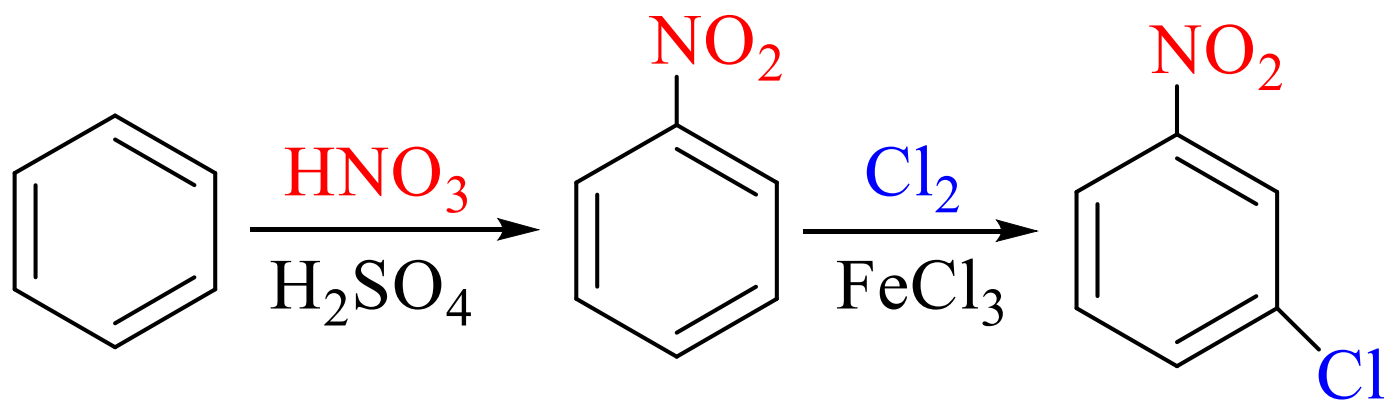


A: (1) 硝化

(2) 氯代

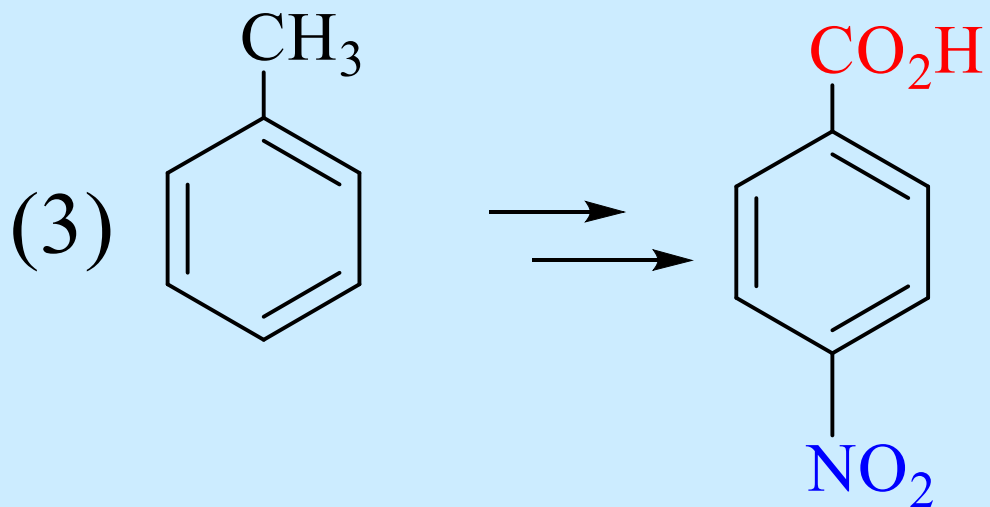
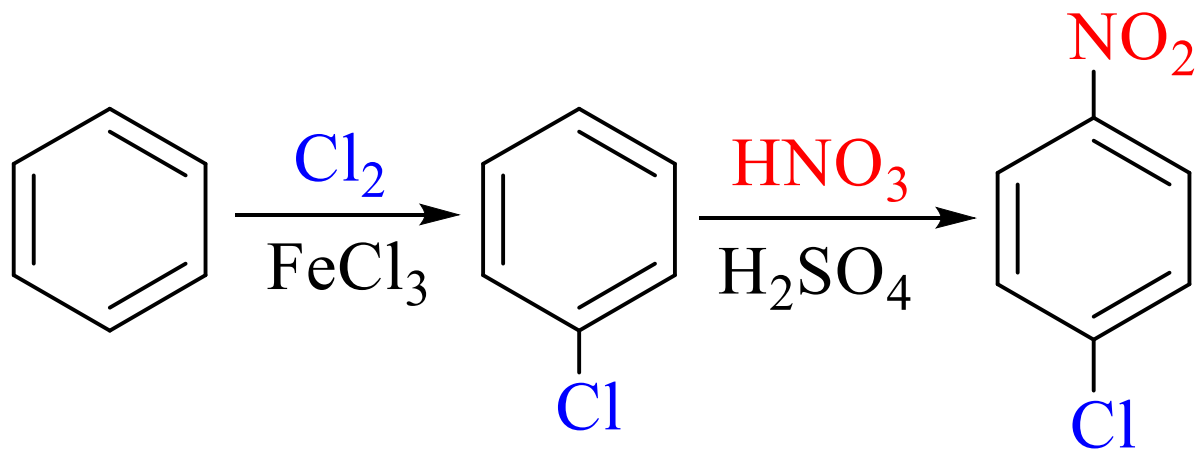
B: (1) 氯代

(2) 硝化

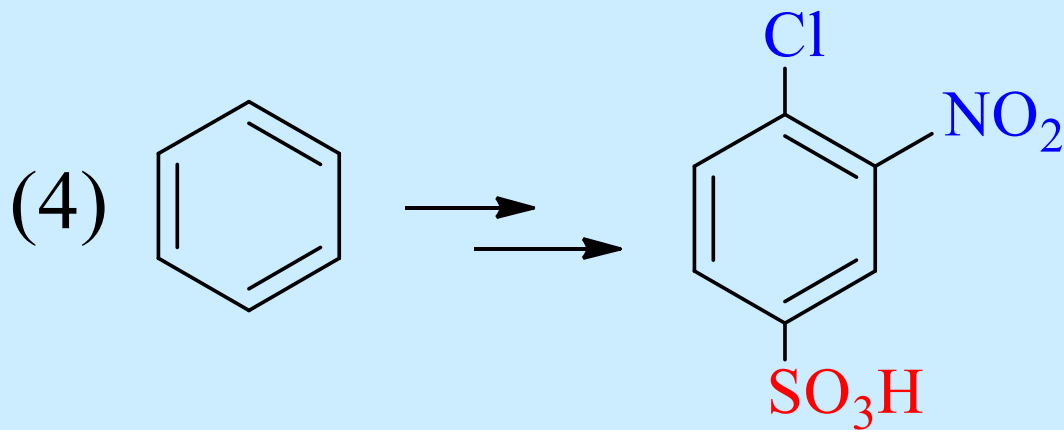
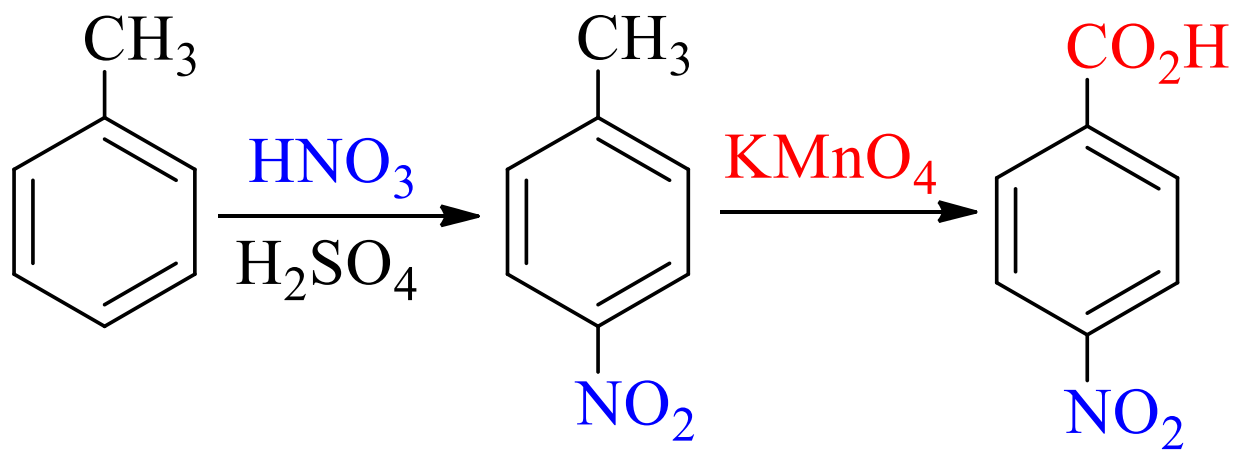


A: (1) 硝化 (2) 氯代

B: (1) 氯代 (2) 硝化



A: (1) 硝化 (2) 氧化  
 B: (1) 氧化 (2) 硝化

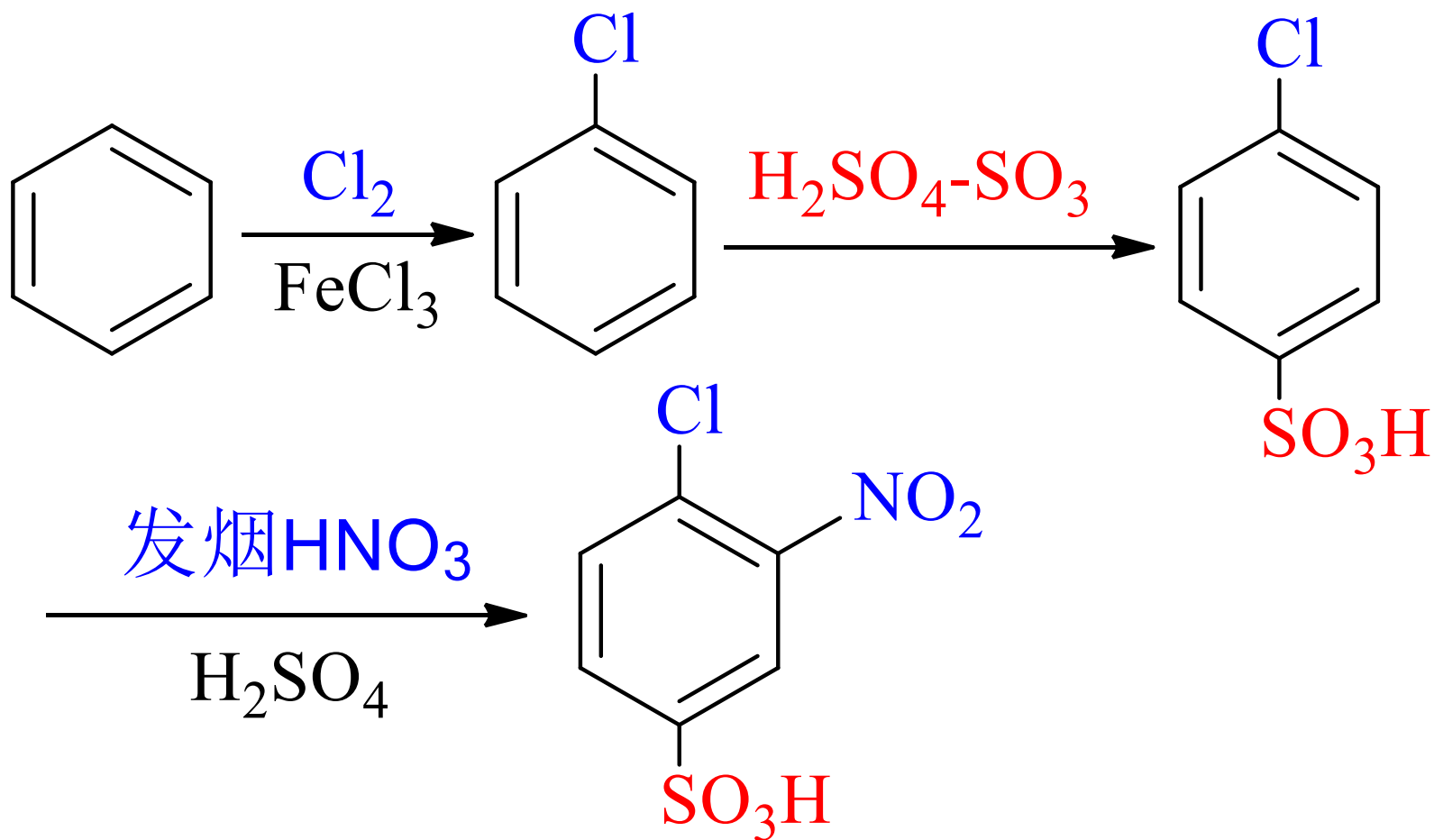


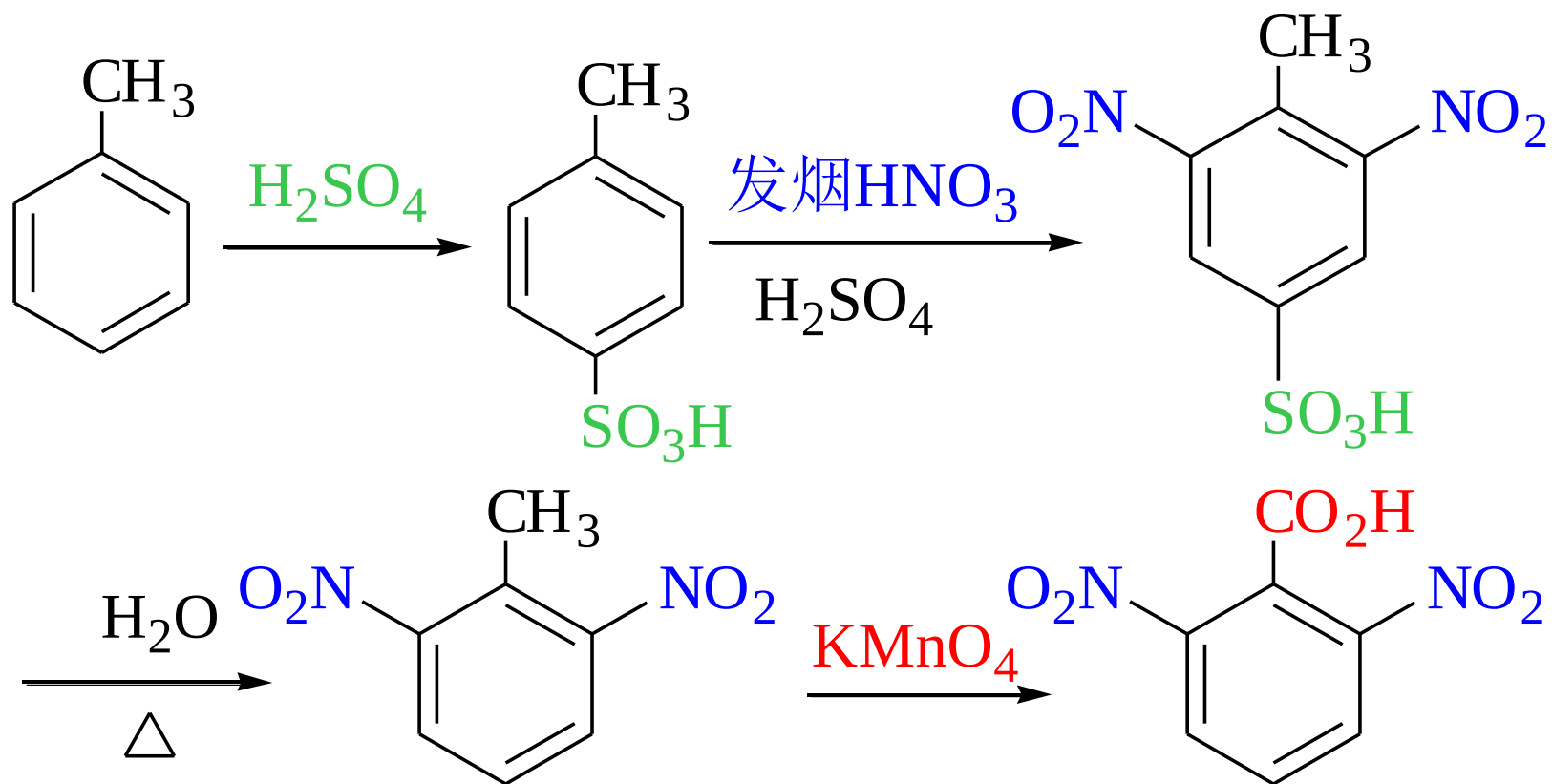
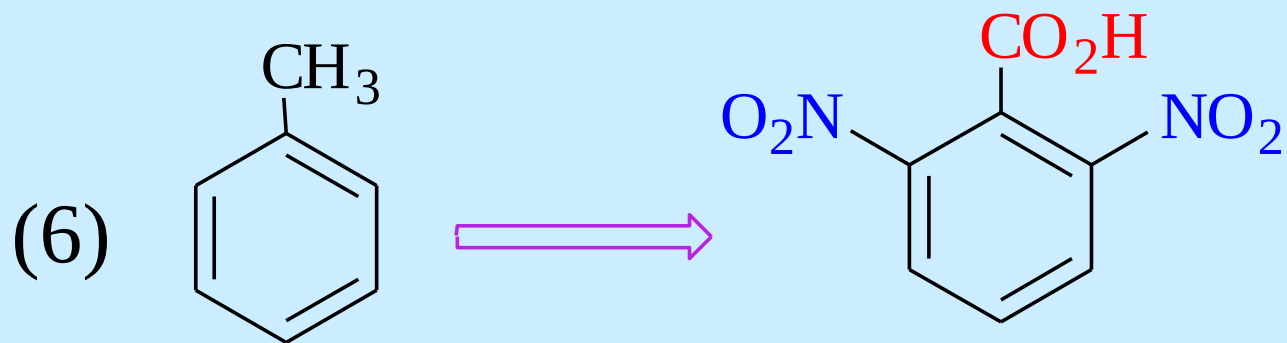
A: (1) 硝化 (2) 氯代 (3) 磺化

B: (1) 氯代 (2) 硝化 (3) 磺化

C: (1) 硝化 (2) 磺化 (3) 氯代

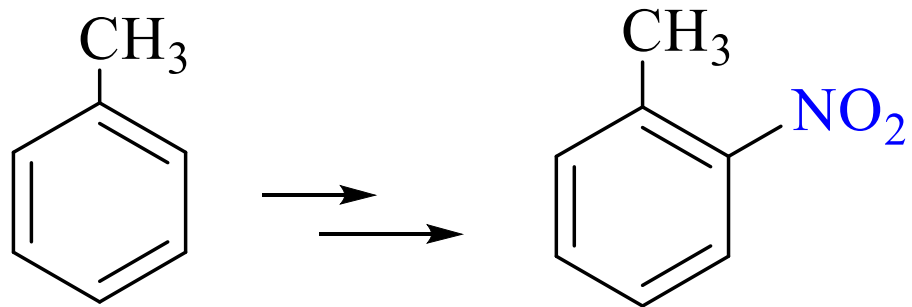
D: (1) 氯代 (2) 磺化 (3) 硝化





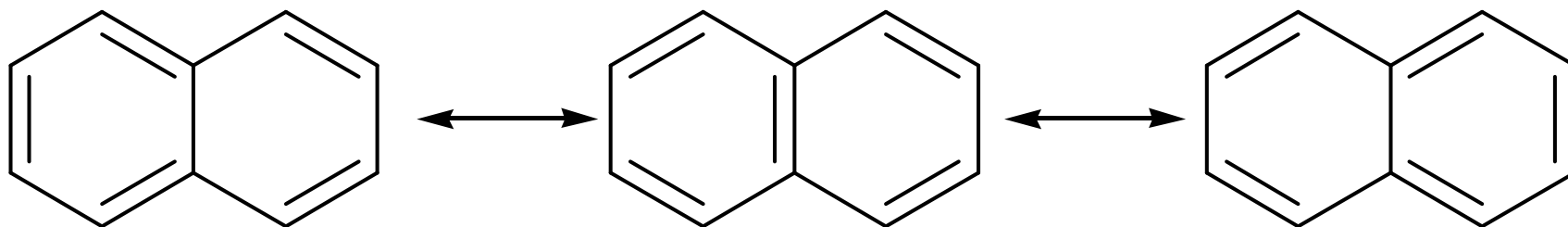
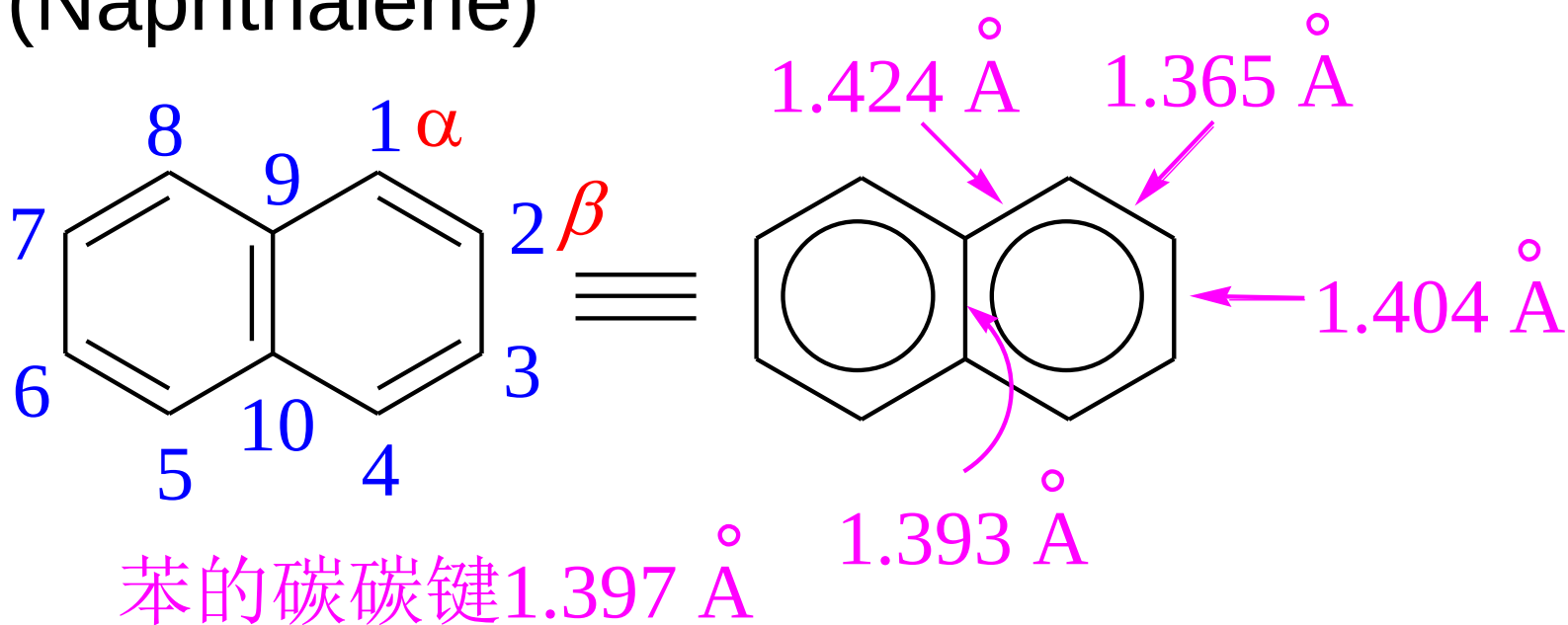
# 测试

完成下列转化



# 稠环芳烃

## 1. 萘 (Naphthalene)

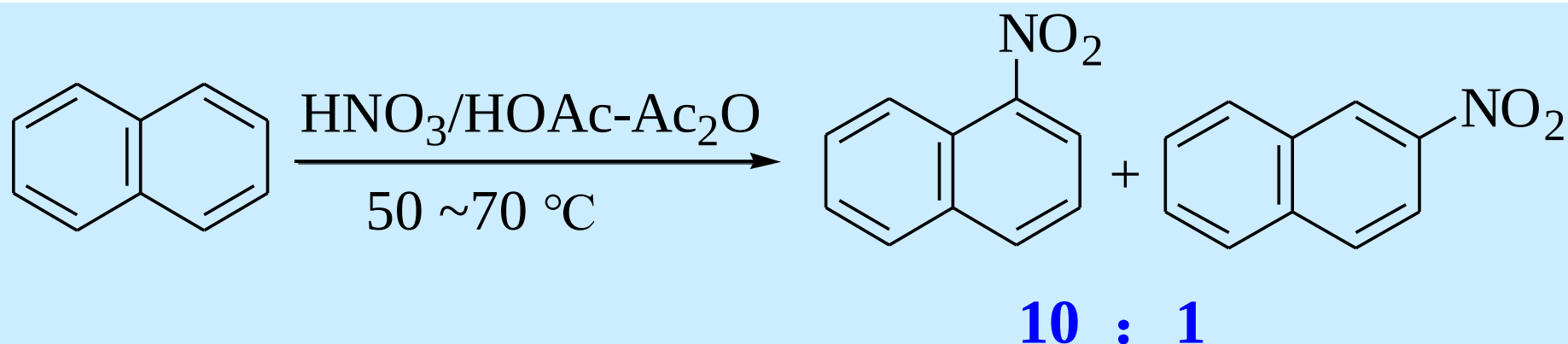


稳定化能 250 kJ/mol

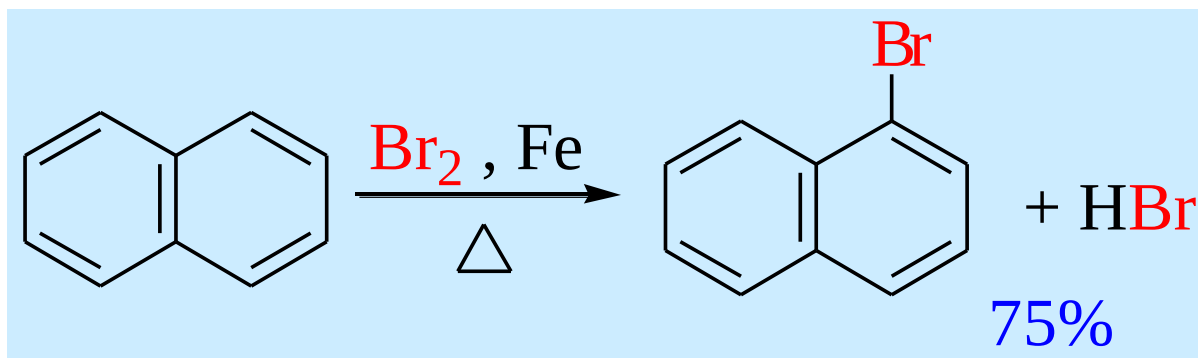


## 2. 萘的反应—亲电取代反应

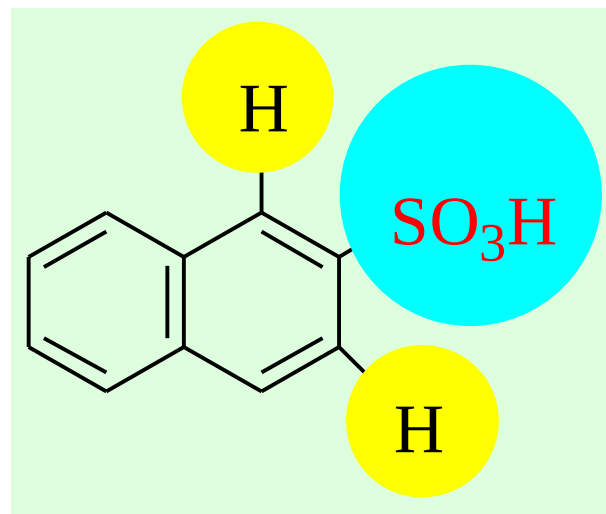
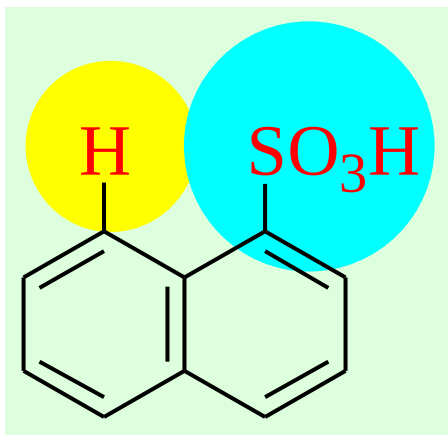
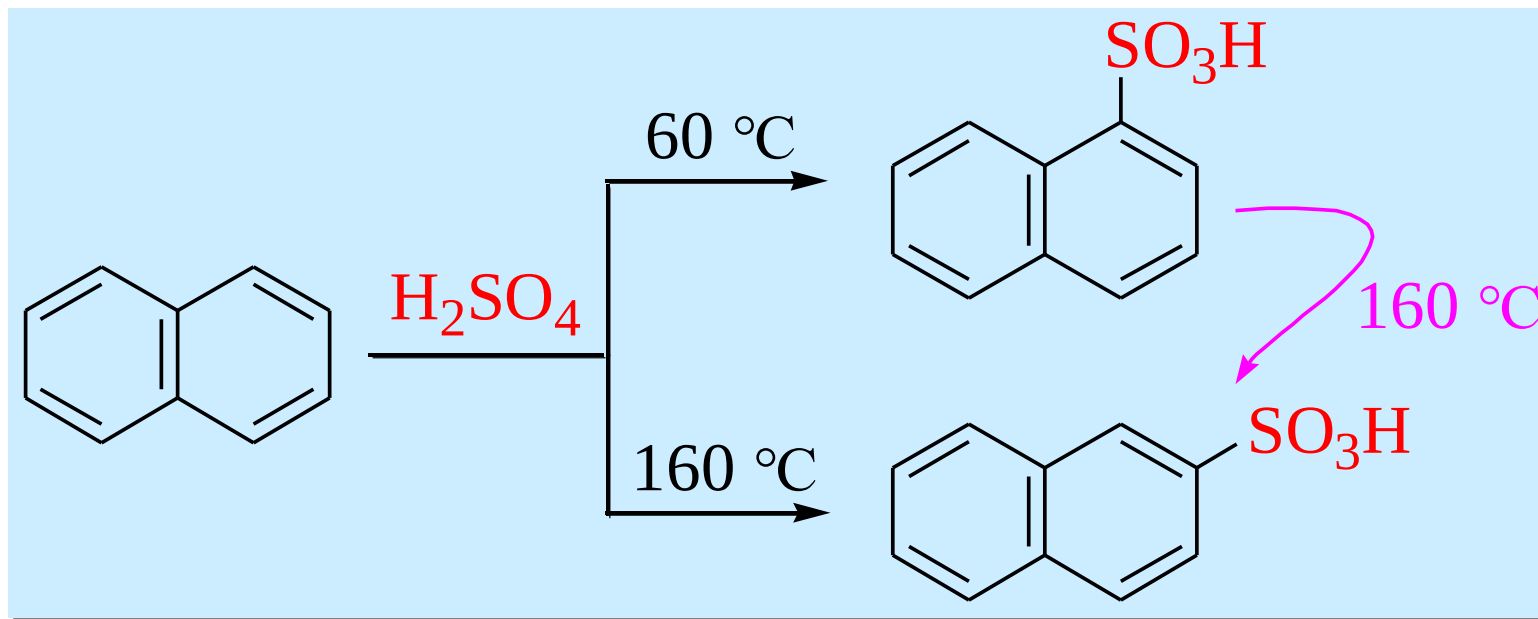
### • 硝化



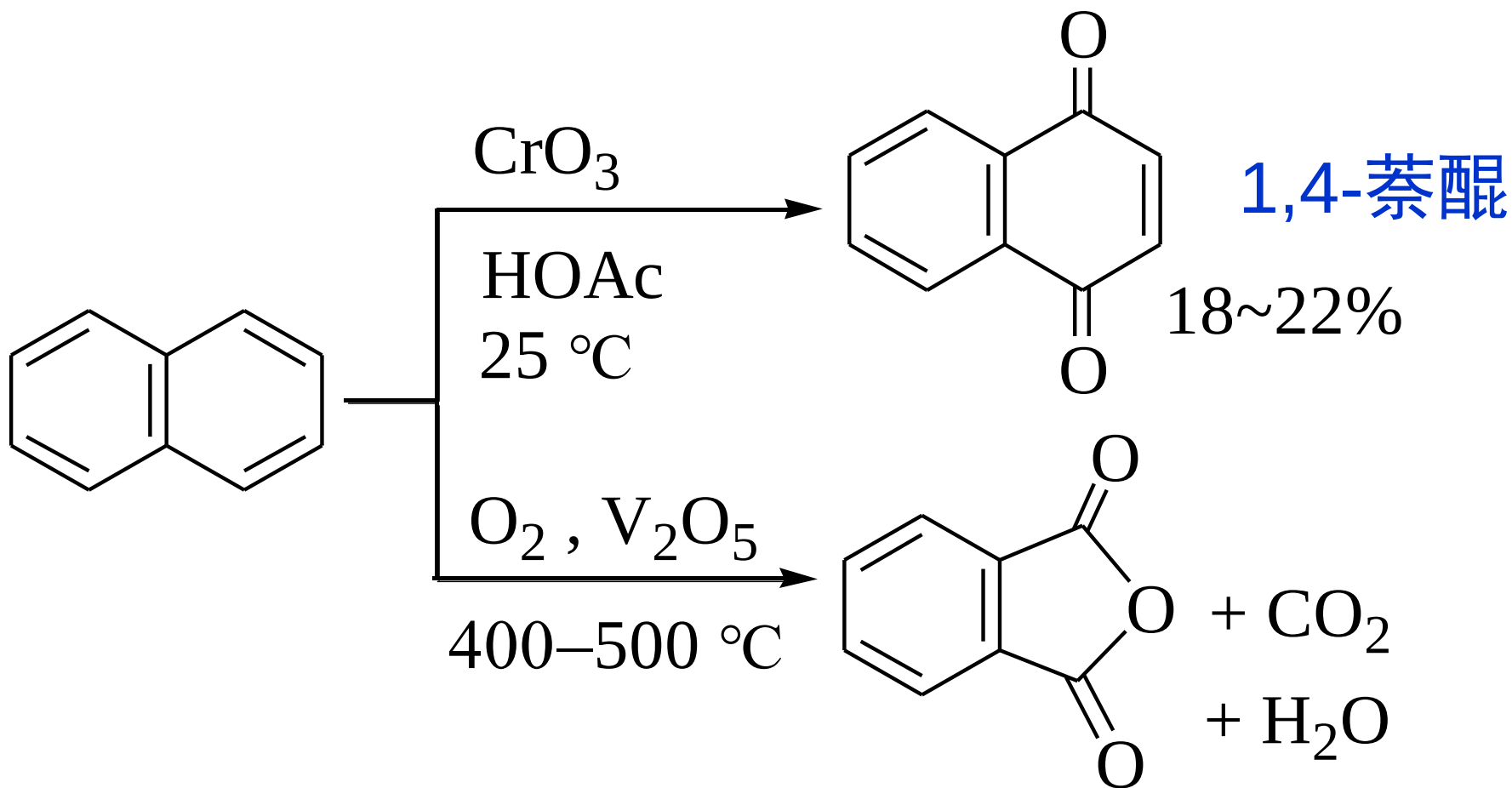
### • 卤代



# • 磺化

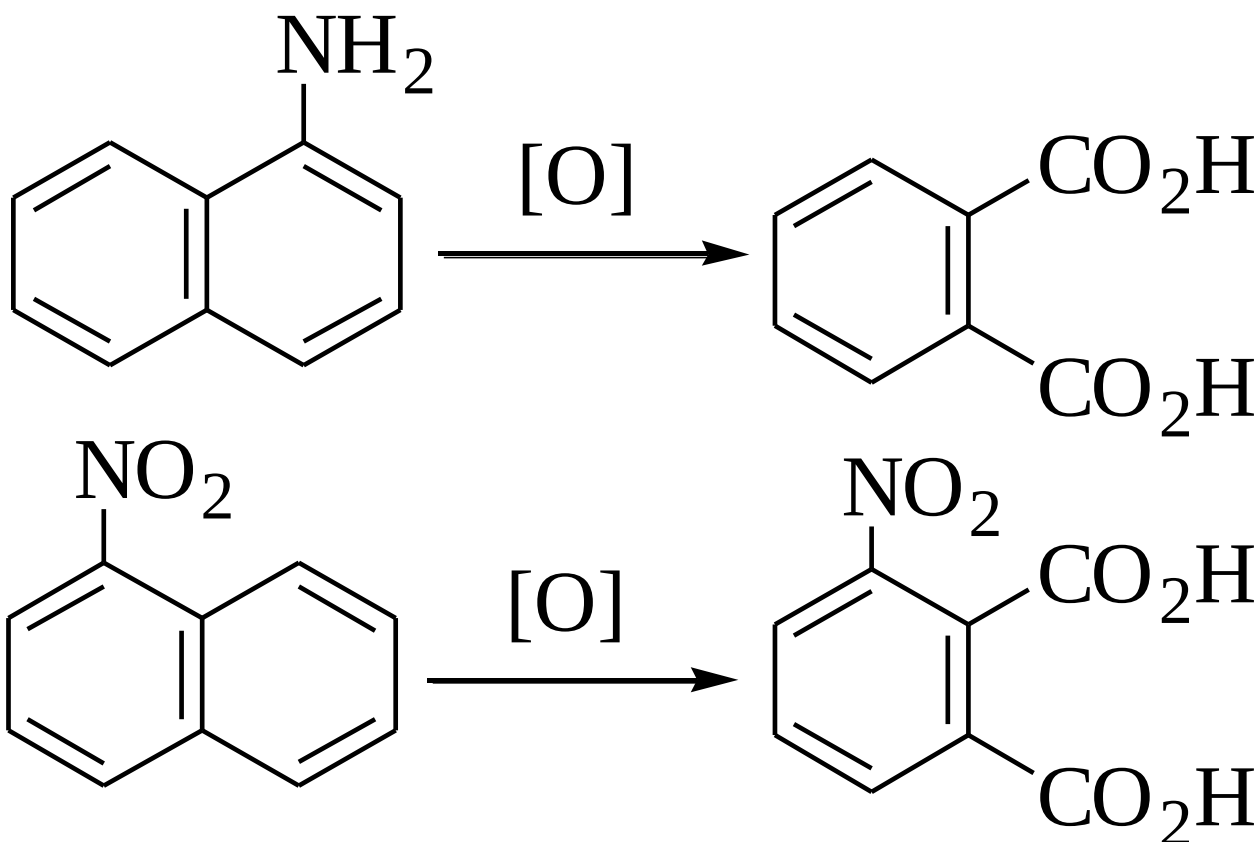


### 3. 萘的反应——氧化反应



邻苯二甲酸酐（苯酐）

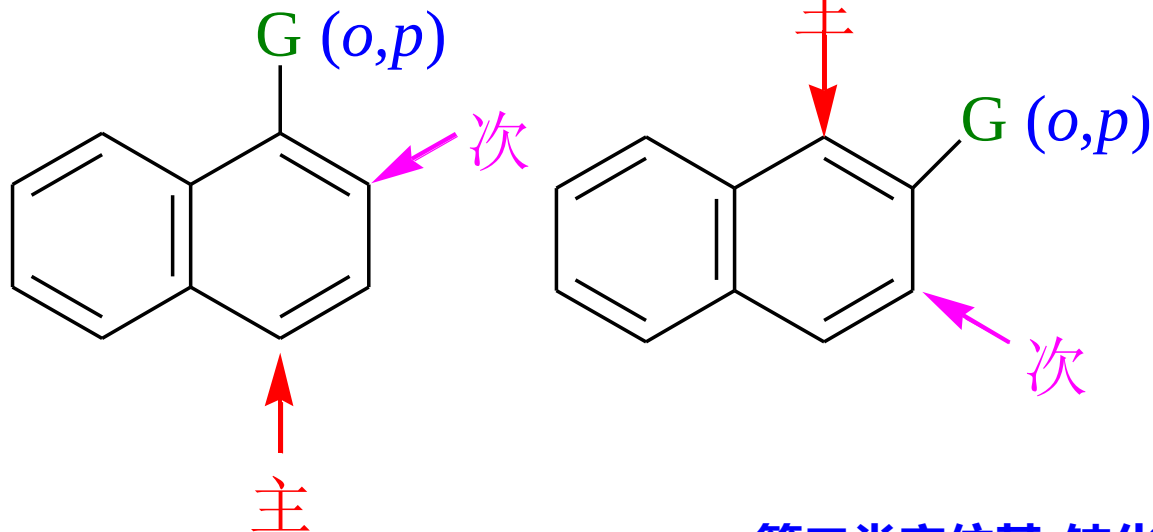
## 4. 取代萘的氧化反应



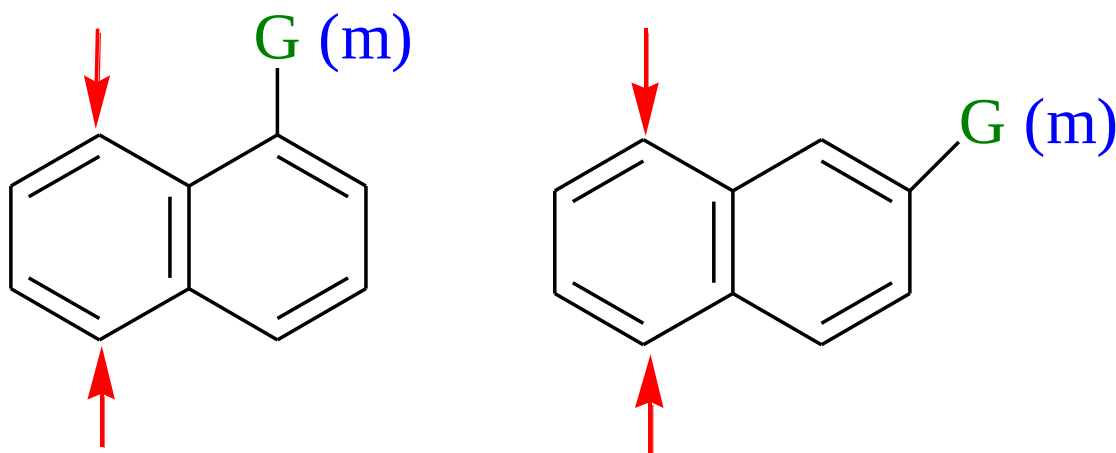
电子云密度较高的环被氧化

# 5. 取代萘亲电取代反应的定位

## 第一类定位基-活化-同环取代

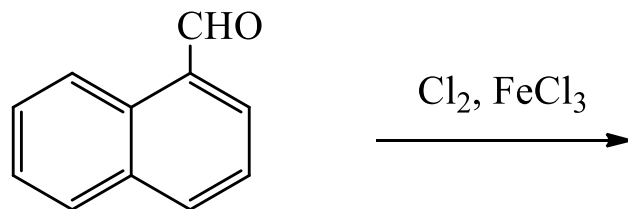
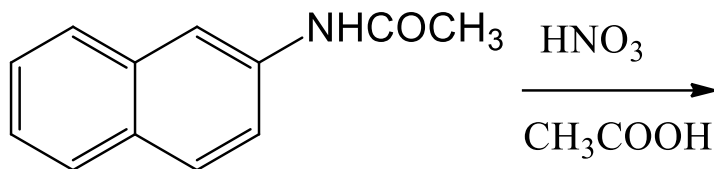
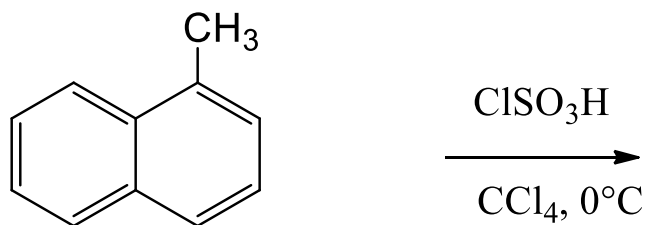


## 第二类定位基-钝化-异环取代



# 测试

完成下列转化



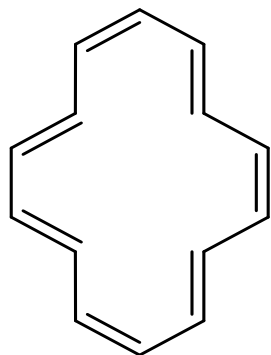
# 芳香性规则

休克尔

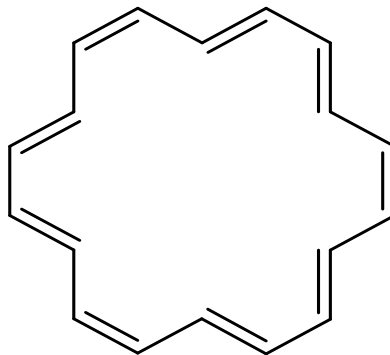
## Hückel's Rule

- 共平面
- 共轭体系
- $(4n + 2) \pi$  electrons,  $n$  为整数
- 具有以上特点的物质具有特殊的稳定性，也就是芳香性。

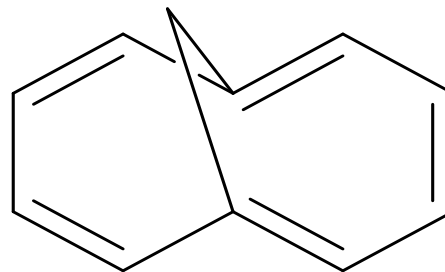
# The Annulenes (轮烯)



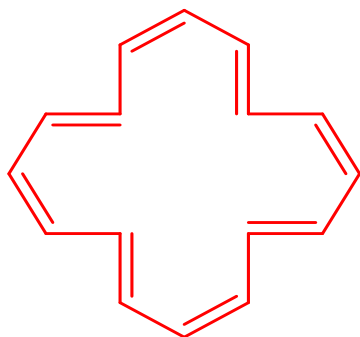
[14]Annulene  
(Aromatic)



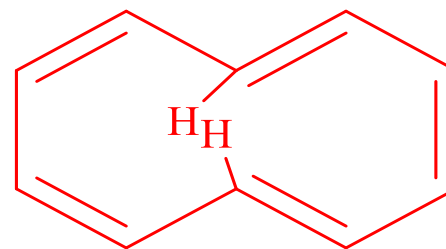
[18]Annulene  
(Aromatic)



(Aromatic)  
(共轭体系共平面)



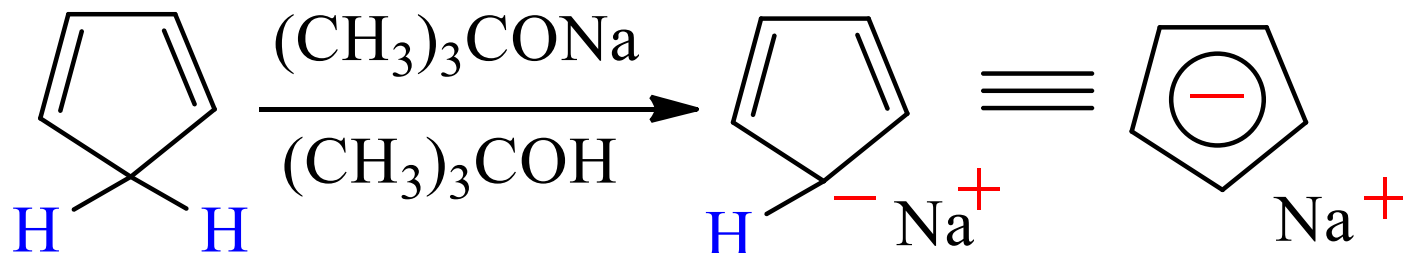
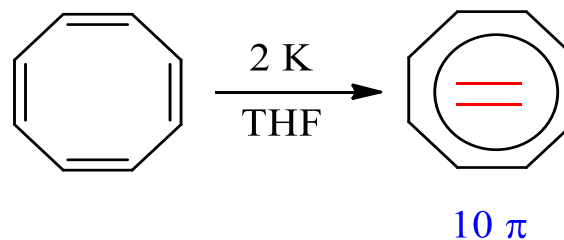
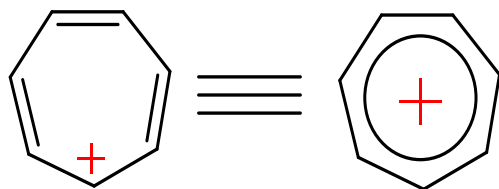
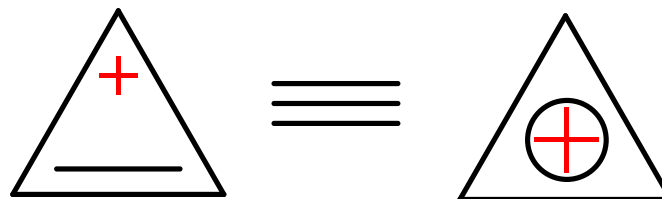
[16]Annulene  
(Not aromatic  
不满足 $4n+2$ )



[10]Annulene  
(Not aromatic  
H与H的位阻,  
不能共平面)

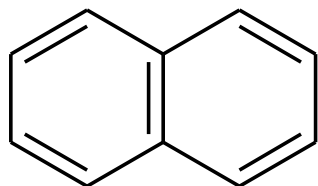


# Aromatic Ions



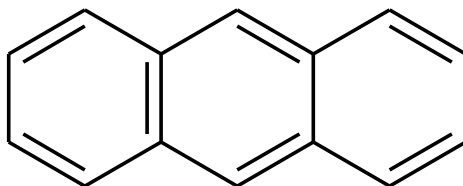
# Other Aromatic Compounds

## Benzenoid Aromatic Compounds



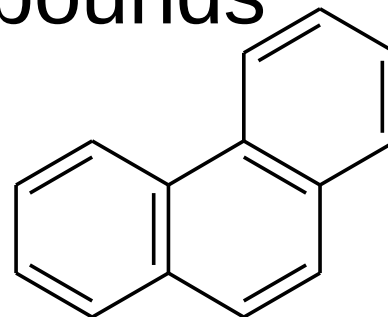
Naphthalene

萘



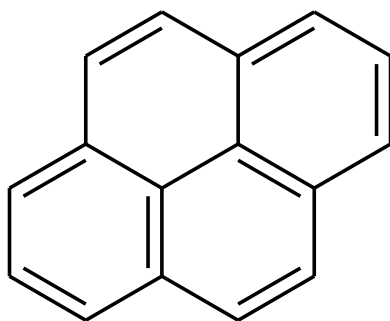
Anthracene

蒽



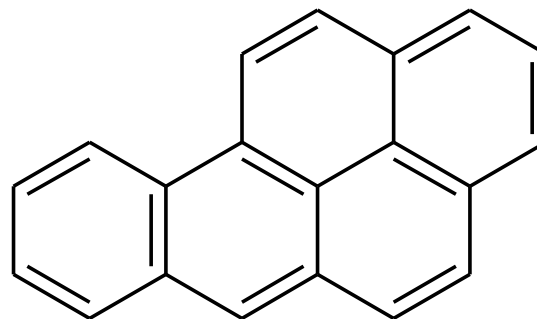
Phenanthrene

菲



Pyrene

芘

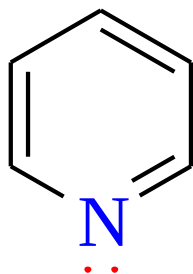


Benzo[α]pyrene

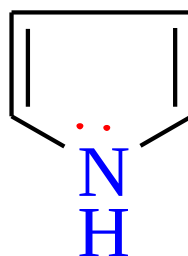
苯并芘

# Heterocyclic Aromatic Compounds:

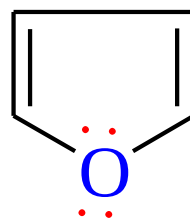
- Heterocyclic compounds containing N, O or S are by far the most common.



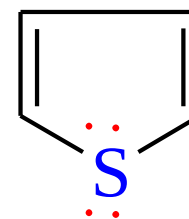
Pyridine  
吡啶



Pyrrole  
吡咯



Furan  
呋喃



Thiophene  
噻吩